Chapter 11

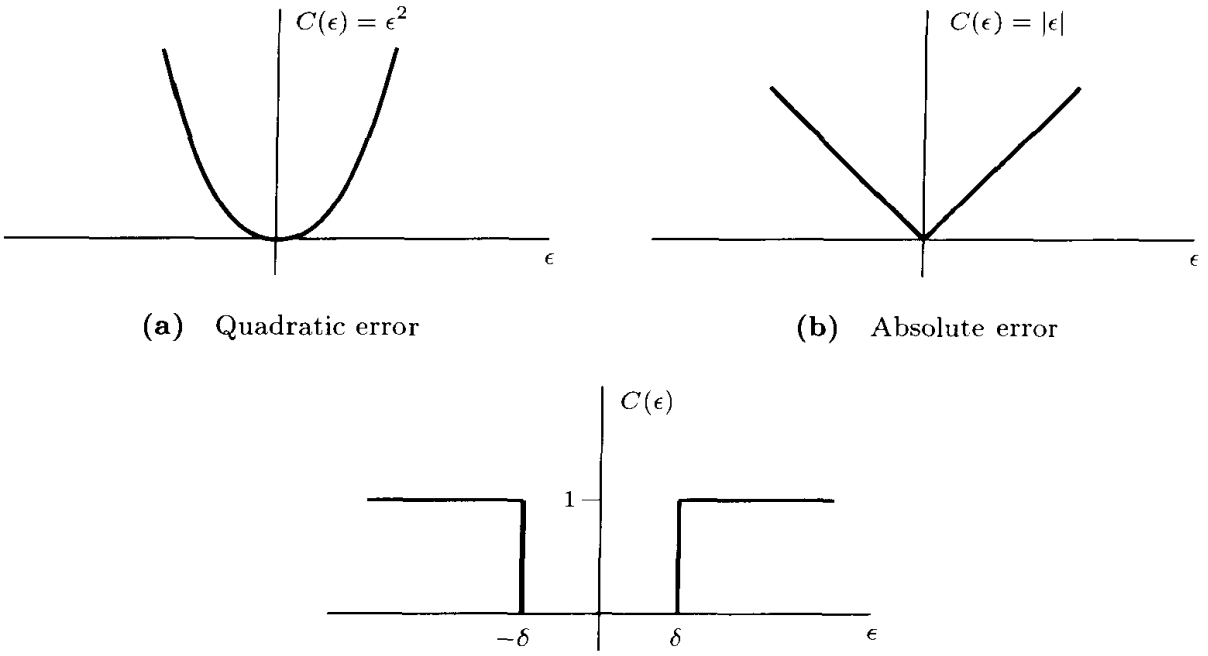
General Bayesian Estimators

11.1 Introduction

Having introduced the Bayesian approach to parameter estimation in the last chapter, we now study more general Bayesian estimators and their properties. To do so, the concept of the Bayesian risk function is discussed. Minimization of this criterion results in a variety of estimators. The ones that we will concentrate on are the MMSE estimator and the maximum a posteriori estimator. These two estimators are the principal ones used in practice. Also, the performance of Bayesian estimators is discussed, leading to the concept of error ellipses. The use of Bayesian estimators in signal processing is illustrated by a deconvolution problem. As a special case we consider in detail the noise filtering problem, in which we use the MMSE estimator in conjunction with the Bayesian linear model to yield the important Wiener filter. In the succeeding chapter we will describe more fully the properties and extensions of the Wiener filter.

11.2 Summary

The Bayes risk is defined in (11.1). For a quadratic cost function the mean of the posterior PDF or the usual MMSE estimator minimizes the risk. A proportional cost function (11.2) results in an optimal estimator which is the median of the posterior PDF. For a "hit-or-miss" cost function (11.3) the mode or maximum location of the posterior PDF is the optimal estimator. The latter is termed the maximum a posteriori (MAP) estimator. The MMSE estimator for a vector parameter is given in (11.10), and the corresponding minimum Bayesian MSE by (11.12). Some examples of the computation of the MAP estimator are included in Section 11.5, followed by the definition for a vector MAP estimator, (11.23) or (11.24). The vector MAP estimator is not a simple extension of the scalar MAP estimator but minimizes a slightly different Bayes risk. It is pointed out that the MMSE and MAP estimators commute over linear transformations, but this property does not carry over to nonlinear ones. The performance of the MMSE estimator is characterized by the PDF of its error (11.26). For the Bayesian linear model this may be determined explicitly from (11.29) and (11.30) and leads to the concept of



(a) Quadratic error

(b) Absolute error

(c) Hit-or-miss error

Figure 11.1 Examples of cost function

an error ellipse as discussed in Example 11.7. Finally, Theorem 11.1 summarizes the MMSE estimator and its performance for the important Bayesian linear model.

11.3 Risk Functions

Previously, we had derived the MMSE estimator by minimizing $E[(\theta - \hat{\theta})^2]$, where the expectation is with respect to the PDF $p(\mathbf{x}, \theta)$. If we let $\epsilon = \theta - \hat{\theta}$ denote the error of the estimator for a particular realization of $\mathbf{x}$ and θ, and also let $C(\epsilon) = \epsilon^2$, then the MSE criterion minimizes $E[C(\epsilon)]$. The deterministic function $C(\epsilon)$ as shown in Figure 11.1a is termed the *cost function*. It is noted that large errors are particularly costly. Also, the average cost or $E[C(\epsilon)]$ is termed the *Bayes risk* $\mathcal{R}$ or

$$\mathcal{R} = E[C(\epsilon)] \tag{11.1}$$

and measures the performance of a given estimator. If $C(\epsilon) = \epsilon^2$, then the cost function is quadratic and the Bayes risk is just the MSE. Of course, there is no need to restrict ourselves to quadratic cost functions, although from a mathematical tractibility standpoint, they are highly desirable. Other possible cost functions are shown in Figures 11.1b and 11.1c. In Figure 11.1b we have

$$C(\epsilon) = |\epsilon|. \tag{11.2}$$

This cost function penalizes errors proportionally. In Figure 11.1c the "hit-or-miss" cost function is displayed. It assigns no cost for small errors and a cost of 1 for all

errors in excess of a threshold error or

$$C(\epsilon) = \begin{cases} 0 & |\epsilon| < \delta \\ 1 & |\epsilon| > \delta \end{cases} \tag{11.3}$$

where $\delta > 0$. If δ is small, we can think of this cost function as assigning the same penalty for any error (a "miss") and no penalty for no error (a "hit"). Note that in all three cases the cost function is symmetric in ϵ, reflecting the implicit assumption that positive errors are just as bad as negative errors. Of course, in general this need not be the case.

We already have seen that the Bayes risk is minimized for a quadratic cost function by the MMSE estimator $\hat{\theta} = E(\theta|\mathbf{x})$. We now determine the optimal estimators for the other cost functions. The Bayes risk $\mathcal{R}$ is

$$\begin{aligned} \mathcal{R} &= E[C(\epsilon)] \\ &= \int\int C(\theta - \hat{\theta}) p(\mathbf{x}, \theta)\, d\mathbf{x}\, d\theta \\ &= \int \left[\int C(\theta - \hat{\theta}) p(\theta|\mathbf{x})\, d\theta \right] p(\mathbf{x})\, d\mathbf{x}. \end{aligned} \tag{11.4}$$

As we did for the MMSE case in Chapter 10, we will attempt to minimize the inner integral for each $\mathbf{x}$. By holding $\mathbf{x}$ fixed $\hat{\theta}$ becomes a scalar variable. First, considering the absolute error cost function, we have for the inner integral of (11.4)

$$\begin{aligned} g(\hat{\theta}) &= \int |\theta - \hat{\theta}| p(\theta|\mathbf{x})\, d\theta \\ &= \int_{-\infty}^{\hat{\theta}} (\hat{\theta} - \theta) p(\theta|\mathbf{x})\, d\theta + \int_{\hat{\theta}}^{\infty} (\theta - \hat{\theta}) p(\theta|\mathbf{x})\, d\theta. \end{aligned}$$

To differentiate with respect to $\hat{\theta}$ we make use of Leibnitz's rule:

$$\begin{aligned} \frac{\partial}{\partial u} \int_{\phi_1(u)}^{\phi_2(u)} h(u, v)\, dv &= \int_{\phi_1(u)}^{\phi_2(u)} \frac{\partial h(u, v)}{\partial u}\, dv + \frac{d\phi_2(u)}{du} h(u, \phi_2(u)) \\ &\quad - \frac{d\phi_1(u)}{du} h(u, \phi_1(u)). \end{aligned}$$

Letting $h(\hat{\theta}, \theta) = (\hat{\theta} - \theta) p(\theta|\mathbf{x})$ for the first integral, we have

$$h(u, \phi_2(u)) = h(\hat{\theta}, \hat{\theta}) = (\hat{\theta} - \hat{\theta}) p(\hat{\theta}|\mathbf{x}) = 0$$

and $d\phi_1(u)/du = 0$ since the lower limit does not depend on u. Similarly, for the second integral the corresponding terms are zero. Hence, we can differentiate the integrand *only* to yield

$$\frac{dg(\hat{\theta})}{d\hat{\theta}} = \int_{-\infty}^{\hat{\theta}} p(\theta|\mathbf{x})\, d\theta - \int_{\hat{\theta}}^{\infty} p(\theta|\mathbf{x})\, d\theta = 0$$

or

$$\int_{-\infty}^{\hat{\theta}} p(\theta|\mathbf{x})\,d\theta = \int_{\hat{\theta}}^{\infty} p(\theta|\mathbf{x})\,d\theta.$$

By definition $\hat{\theta}$ is the *median* of the posterior PDF or the point for which $\Pr\{\theta \leq \hat{\theta}|\mathbf{x}\} = 1/2$.

For the "hit-or-miss" cost function we have $\mathcal{C}(\epsilon) = 1$ for $\epsilon > \delta$ and $\epsilon < -\delta$ or for $\theta > \hat{\theta} + \delta$ and $\theta < \hat{\theta} - \delta$, so that the inner integral in (11.4) is

$$g(\hat{\theta}) = \int_{-\infty}^{\hat{\theta}-\delta} 1 \cdot p(\theta|\mathbf{x})\,d\theta + \int_{\hat{\theta}+\delta}^{\infty} 1 \cdot p(\theta|\mathbf{x})\,d\theta.$$

But

$$\int_{-\infty}^{\infty} p(\theta|\mathbf{x})\,d\theta = 1,$$

yielding

$$g(\hat{\theta}) = 1 - \int_{\hat{\theta}-\delta}^{\hat{\theta}+\delta} p(\theta|\mathbf{x})\,d\theta.$$

This is minimized by maximizing

$$\int_{\hat{\theta}-\delta}^{\hat{\theta}+\delta} p(\theta|\mathbf{x})\,d\theta.$$

For δ arbitrarily small this is maximized by choosing $\hat{\theta}$ to correspond to the location of the *maximum* of $p(\theta|\mathbf{x})$. The estimator that minimizes the Bayes risk for the "hit-or-miss" cost function is therefore the *mode* (location of the maximum) of the posterior PDF. It is termed the *maximum a posteriori* (MAP) estimator and will be described in more detail later.

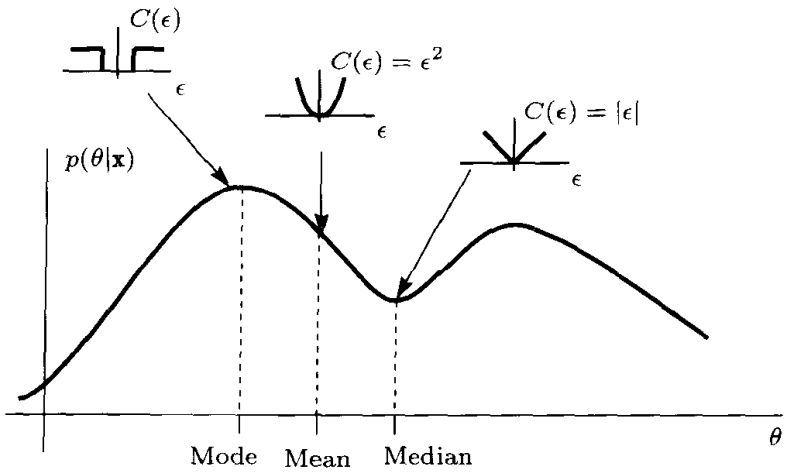
In summary, the estimators that minimize the Bayes risk for the cost functions of Figure 11.1 are the *mean*, *median*, and *mode* of the posterior PDF. This is illustrated in Figure 11.2a. For some posterior PDFs these three estimators are identical. A notable example is the Gaussian posterior PDF

$$p(\theta|\mathbf{x}) = \frac{1}{\sqrt{2\pi\sigma_{\theta|x}^2}} \exp\left[-\frac{1}{2\sigma_{\theta|x}^2}(\theta - \mu_{\theta|x})^2\right].$$

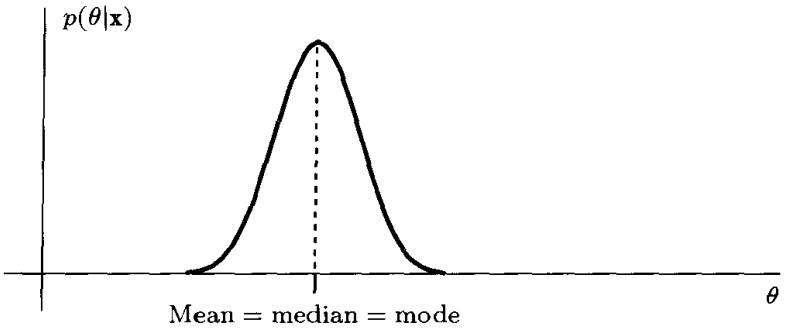
The mean $\mu_{\theta|x}$ is identical to the median (due to the symmetry) and the mode, as illustrated in Figure 11.2b. (See also Problem 11.2.)

11.4 Minimum Mean Square Error Estimators

In Chapter 10 the MMSE estimator was determined to be $E(\theta|\mathbf{x})$ or the mean of the posterior PDF. For this reason it is also commonly referred to as the *conditional mean*



(a) General posterior PDF



(b) Gaussian posterior PDF

Figure 11.2 Estimators for different cost functions

estimator. We continue our discussion of this important estimator by first extending it to the vector parameter case and then studying some of its properties.

If $\boldsymbol{\theta}$ is a vector parameter of dimension $p \times 1$, then to estimate θ_1, for example, we may view the remaining parameters as nuisance parameters (see Chapter 10). If $p(\mathbf{x}|\boldsymbol{\theta})$ is the conditional PDF of the data and $p(\boldsymbol{\theta})$ the prior PDF of the vector parameter, we may obtain the posterior PDF for θ_1 as

$$p(\theta_1|\mathbf{x}) = \int \cdots \int p(\boldsymbol{\theta}|\mathbf{x})\, d\theta_2 \cdots d\theta_p \tag{11.5}$$

where

$$p(\boldsymbol{\theta}|\mathbf{x}) = \frac{p(\mathbf{x}|\boldsymbol{\theta})p(\boldsymbol{\theta})}{\int p(\mathbf{x}|\boldsymbol{\theta})p(\boldsymbol{\theta})\, d\boldsymbol{\theta}}. \tag{11.6}$$

Then, by the same reasoning as in Chapter 10 we have

$$\begin{aligned}\hat{\theta}_1 &= E(\theta_1|\mathbf{x}) \\ &= \int \theta_1 p(\theta_1|\mathbf{x})\, d\theta_1\end{aligned}$$

or in general

$$\hat{\theta}_i = \int \theta_i p(\theta_i|\mathbf{x})\, d\theta_i \qquad i = 1, 2, \ldots, p. \tag{11.7}$$

This is the MMSE estimator that minimizes

$$E[(\theta_i - \hat{\theta}_i)^2] = \int (\theta_i - \hat{\theta}_i)^2 p(\mathbf{x}, \theta_i)\, d\mathbf{x}\, d\theta_i \tag{11.8}$$

or the squared error when averaged with respect to the *marginal* PDF $p(\mathbf{x}, \theta_i)$. Thus, the MMSE estimator for a vector parameter does not entail anything new but only a need to determine the posterior PDF for each parameter. Alternatively, we can express the MMSE estimator for the first parameter from (11.5) as

$$\begin{aligned}\hat{\theta}_1 &= \int \theta_1 p(\theta_1|\mathbf{x})\, d\theta_1 \\ &= \int \theta_1 \left[\int \cdots \int p(\boldsymbol{\theta}|\mathbf{x})\, d\theta_2 \ldots d\theta_p \right] d\theta_1 \\ &= \int \theta_1 p(\boldsymbol{\theta}|\mathbf{x})\, d\boldsymbol{\theta}\end{aligned}$$

or in general

$$\hat{\theta}_i = \int \theta_i p(\boldsymbol{\theta}|\mathbf{x})\, d\boldsymbol{\theta} \qquad i = 1, 2, \ldots, p.$$

In vector form we have

$$\begin{aligned}\hat{\boldsymbol{\theta}} &= \begin{bmatrix} \int \theta_1 p(\boldsymbol{\theta}|\mathbf{x})\, d\boldsymbol{\theta} \\ \int \theta_2 p(\boldsymbol{\theta}|\mathbf{x})\, d\boldsymbol{\theta} \\ \vdots \\ \int \theta_p p(\boldsymbol{\theta}|\mathbf{x})\, d\boldsymbol{\theta} \end{bmatrix} \\ &= \int \boldsymbol{\theta} p(\boldsymbol{\theta}|\mathbf{x})\, d\boldsymbol{\theta} \qquad (11.9) \\ &= E(\boldsymbol{\theta}|\mathbf{x}) \qquad (11.10)\end{aligned}$$

where the expectation is with respect to the posterior PDF of the *vector parameter* or $p(\boldsymbol{\theta}|\mathbf{x})$. Note that the vector MMSE estimator $E(\boldsymbol{\theta}|\mathbf{x})$ *minimizes the MSE for each component of the unknown vector parameter*, or $[\hat{\boldsymbol{\theta}}]_i = [E(\boldsymbol{\theta}|\mathbf{x})]_i$ minimizes $E[(\theta_i - \hat{\theta}_i)^2]$. This follows from the derivation.

As discussed in Chapter 10, the minimum Bayesian MSE for a scalar parameter is the posterior PDF variance when averaged over the PDF of $\mathbf{x}$ (see (10.13)). This is because

$$\text{Bmse}(\hat{\theta}_1) = E[(\theta_1 - \hat{\theta}_1)^2] = \int (\theta_1 - \hat{\theta}_1)^2 p(\mathbf{x}, \theta_1)\, d\theta_1 d\mathbf{x}$$

and since $\hat{\theta}_1 = E(\theta_1|\mathbf{x})$, we have

$$\begin{aligned} \text{Bmse}(\hat{\theta}_1) &= \int \left[\int (\theta_1 - E(\theta_1|\mathbf{x}))^2 p(\theta_1|\mathbf{x})\, d\theta_1 \right] p(\mathbf{x})\, d\mathbf{x} \\ &= \int \text{var}(\theta_1|\mathbf{x}) p(\mathbf{x})\, d\mathbf{x}. \end{aligned}$$

Now, however, the posterior PDF can be written as

$$p(\theta_1|\mathbf{x}) = \int \cdots \int p(\boldsymbol{\theta}|\mathbf{x})\, d\theta_2 \ldots d\theta_p$$

so that

$$\text{Bmse}(\hat{\theta}_1) = \int \left[\int (\theta_1 - E(\theta_1|\mathbf{x}))^2 p(\boldsymbol{\theta}|\mathbf{x})\, d\boldsymbol{\theta} \right] p(\mathbf{x})\, d\mathbf{x}. \tag{11.11}$$

The inner integral in (11.11) is the variance of θ_1 for the posterior PDF $p(\boldsymbol{\theta}|\mathbf{x})$. This is just the [1,1] element of $\mathbf{C}_{\theta|x}$, the covariance matrix of the posterior PDF. Hence, in general we have that the minimum Bayesian MSE is

$$\text{Bmse}(\hat{\theta}_i) = \int [\mathbf{C}_{\theta|x}]_{ii} p(\mathbf{x})\, d\mathbf{x} \qquad i = 1, 2, \ldots, p \tag{11.12}$$

where

$$\mathbf{C}_{\theta|x} = E_{\theta|x}\left[(\boldsymbol{\theta} - E(\boldsymbol{\theta}|\mathbf{x}))(\boldsymbol{\theta} - E(\boldsymbol{\theta}|\mathbf{x}))^T \right]. \tag{11.13}$$

An example follows.

Example 11.1 - Bayesian Fourier Analysis

We reconsider Example 4.2, but to simplify the calculations we let $M = 1$ so that our data model becomes

$$x[n] = a \cos 2\pi f_0 n + b \sin 2\pi f_0 n + w[n] \qquad n = 0, 1, \ldots, N-1$$

where f_0 is a multiple of $1/N$, excepting 0 or $1/2$ (for which $\sin 2\pi f_0 n$ is identically zero), and $w[n]$ is WGN with variance σ^2. It is desired to estimate $\boldsymbol{\theta} = [a\, b]^T$. We depart from the classical model by assuming a, b are random variables with prior PDF

$$\boldsymbol{\theta} \sim \mathcal{N}(\mathbf{0}, \sigma_\theta^2 \mathbf{I})$$

and $\boldsymbol{\theta}$ is independent of $w[n]$. This type of model is referred to as a *Rayleigh fading sinusoid* [Van Trees 1968] and is frequently used to represent a sinusoid that has

propagated through a dispersive medium (see also Problem 11.6). To find the MMSE estimator we need to evaluate $E(\boldsymbol{\theta}|\mathbf{x})$. The data model is rewritten as

$$\mathbf{x} = \mathbf{H}\boldsymbol{\theta} + \mathbf{w}$$

where

$$\mathbf{H} = \begin{bmatrix} 1 & 0 \\ \cos 2\pi f_0 & \sin 2\pi f_0 \\ \vdots & \vdots \\ \cos[2\pi f_0(N-1)] & \sin[2\pi f_0(N-1)] \end{bmatrix}$$

which is recognized as the Bayesian linear model. From Theorem 10.3 we can obtain the mean as well as the covariance of the posterior PDF. To do so we let

$$\begin{aligned} \boldsymbol{\mu}_\theta &= \mathbf{0} \\ \mathbf{C}_\theta &= \sigma_\theta^2 \mathbf{I} \\ \mathbf{C}_w &= \sigma^2 \mathbf{I} \end{aligned}$$

to obtain

$$\begin{aligned} \hat{\boldsymbol{\theta}} &= E(\boldsymbol{\theta}|\mathbf{x}) = \sigma_\theta^2 \mathbf{H}^T (\mathbf{H}\sigma_\theta^2 \mathbf{H}^T + \sigma^2 \mathbf{I})^{-1} \mathbf{x} \\ \mathbf{C}_{\theta|x} &= \sigma_\theta^2 \mathbf{I} - \sigma_\theta^2 \mathbf{H}^T (\mathbf{H}\sigma_\theta^2 \mathbf{H}^T + \sigma^2 \mathbf{I})^{-1} \mathbf{H}\sigma_\theta^2. \end{aligned}$$

A somewhat more convenient form is given by (10.32) and (10.33) as

$$\begin{aligned} \hat{\boldsymbol{\theta}} &= E(\boldsymbol{\theta}|\mathbf{x}) = \left(\frac{1}{\sigma_\theta^2}\mathbf{I} + \mathbf{H}^T \frac{1}{\sigma^2}\mathbf{H}\right)^{-1} \mathbf{H}^T \frac{1}{\sigma^2}\mathbf{x} \\ \mathbf{C}_{\theta|x} &= \left(\frac{1}{\sigma_\theta^2}\mathbf{I} + \mathbf{H}^T \frac{1}{\sigma^2}\mathbf{H}\right)^{-1}. \end{aligned}$$

Now, because the columns of $\mathbf{H}$ are orthogonal (due to the choice of frequency), we have (see Example 4.2)

$$\mathbf{H}^T\mathbf{H} = \frac{N}{2}\mathbf{I}$$

and

$$\begin{aligned} \hat{\boldsymbol{\theta}} &= \left(\frac{1}{\sigma_\theta^2}\mathbf{I} + \frac{N}{2\sigma^2}\mathbf{I}\right)^{-1} \frac{\mathbf{H}^T\mathbf{x}}{\sigma^2} \\ &= \frac{\frac{1}{\sigma^2}}{\frac{1}{\sigma_\theta^2} + \frac{N}{2\sigma^2}} \mathbf{H}^T\mathbf{x} \end{aligned}$$

or the MMSE estimator is

$$\begin{aligned} \hat{a} &= \frac{1}{1 + \frac{2\sigma^2/N}{\sigma_\theta^2}} \left[\frac{2}{N}\sum_{n=0}^{N-1} x[n]\cos 2\pi f_0 n\right] \\ \hat{b} &= \frac{1}{1 + \frac{2\sigma^2/N}{\sigma_\theta^2}} \left[\frac{2}{N}\sum_{n=0}^{N-1} x[n]\sin 2\pi f_0 n\right]. \end{aligned}$$

The results differ from the classical case only in the scale factor, and if $\sigma_\theta^2 \gg 2\sigma^2/N$, the two results are identical. This corresponds to little prior knowledge compared to the data knowledge. The posterior covariance matrix is

$$\mathbf{C}_{\theta|x} = \frac{1}{\frac{1}{\sigma_\theta^2} + \frac{N}{2\sigma^2}}\mathbf{I}$$

which does not depend on $\mathbf{x}$. Hence, from (11.12)

$$\begin{aligned} \text{Bmse}(\hat{a}) &= \frac{1}{\frac{1}{\sigma_\theta^2} + \frac{1}{2\sigma^2/N}} \\ \text{Bmse}(\hat{b}) &= \frac{1}{\frac{1}{\sigma_\theta^2} + \frac{1}{2\sigma^2/N}}. \end{aligned}$$

◇

It is interesting to note that in the absence of prior knowledge in the Bayesian linear model the MMSE estimator yields the same form as the MVU estimator for the classical linear model. Many fortuitous circumstances enter into making this so, as described in Problem 11.7. To verify this result note that from (10.32)

$$\hat{\boldsymbol{\theta}} = E(\boldsymbol{\theta}|\mathbf{x}) = \boldsymbol{\mu}_\theta + (\mathbf{C}_\theta^{-1} + \mathbf{H}^T\mathbf{C}_w^{-1}\mathbf{H})^{-1}\mathbf{H}^T\mathbf{C}_w^{-1}(\mathbf{x} - \mathbf{H}\boldsymbol{\mu}_\theta).$$

For no prior knowledge $\mathbf{C}_\theta^{-1} \to \mathbf{0}$, and therefore,

$$\hat{\boldsymbol{\theta}} \to (\mathbf{H}^T\mathbf{C}_w^{-1}\mathbf{H})^{-1}\mathbf{H}^T\mathbf{C}_w^{-1}\mathbf{x} \tag{11.14}$$

which is recognized as the MVU estimator for the general linear model (see Chapter 4).

The MMSE estimator has several useful properties that will be exploited in our study of Kalman filters in Chapter 13 (see also Problems 11.8 and 11.9). First, it commutes over linear (actually affine) transformations. Assume that we wish to estimate $\boldsymbol{\alpha}$ for

$$\boldsymbol{\alpha} = \mathbf{A}\boldsymbol{\theta} + \mathbf{b} \tag{11.15}$$

where $\mathbf{A}$ is a known $r \times p$ matrix and $\mathbf{b}$ is a known $r \times 1$ vector. Then, $\boldsymbol{\alpha}$ is a random vector for which the MMSE estimator is

$$\hat{\boldsymbol{\alpha}} = E(\boldsymbol{\alpha}|\mathbf{x}).$$

Because of the linearity of the expectation operator

$$\begin{aligned} \hat{\boldsymbol{\alpha}} &= E(\mathbf{A}\boldsymbol{\theta} + \mathbf{b}|\mathbf{x}) \\ &= \mathbf{A}E(\boldsymbol{\theta}|\mathbf{x}) + \mathbf{b} \\ &= \mathbf{A}\hat{\boldsymbol{\theta}} + \mathbf{b}. \end{aligned} \tag{11.16}$$

This holds regardless of the joint PDF $p(\mathbf{x}, \boldsymbol{\theta})$. A second important property focuses on the MMSE estimator based on two data vectors $\mathbf{x}_1, \mathbf{x}_2$. We assume that $\boldsymbol{\theta}, \mathbf{x}_1, \mathbf{x}_2$ are jointly Gaussian and the data vectors are independent. The MMSE estimator is

$$\hat{\boldsymbol{\theta}} = E(\boldsymbol{\theta}|\mathbf{x}_1, \mathbf{x}_2).$$

Letting $\mathbf{x} = [\mathbf{x}_1^T \ \mathbf{x}_2^T]^T$, we have from Theorem 10.2

$$\hat{\boldsymbol{\theta}} = E(\boldsymbol{\theta}|\mathbf{x}) = E(\boldsymbol{\theta}) + \mathbf{C}_{\theta x}\mathbf{C}_{xx}^{-1}(\mathbf{x} - E(\mathbf{x})). \tag{11.17}$$

Since $\mathbf{x}_1, \mathbf{x}_2$ are independent,

$$\begin{aligned} \mathbf{C}_{xx}^{-1} &= \begin{bmatrix} \mathbf{C}_{x_1x_1} & \mathbf{C}_{x_1x_2} \\ \mathbf{C}_{x_2x_1} & \mathbf{C}_{x_2x_2} \end{bmatrix}^{-1} \\ &= \begin{bmatrix} \mathbf{C}_{x_1x_1} & \mathbf{0} \\ \mathbf{0} & \mathbf{C}_{x_2x_2} \end{bmatrix}^{-1} \\ &= \begin{bmatrix} \mathbf{C}_{x_1x_1}^{-1} & \mathbf{0} \\ \mathbf{0} & \mathbf{C}_{x_2x_2}^{-1} \end{bmatrix} \end{aligned}$$

and also

$$\mathbf{C}_{\theta x} = E\left[\boldsymbol{\theta}\begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \end{bmatrix}^T\right] = \begin{bmatrix} \mathbf{C}_{\theta x_1} & \mathbf{C}_{\theta x_2} \end{bmatrix}.$$

It follows from (11.17) that

$$\begin{aligned} \hat{\boldsymbol{\theta}} &= E(\boldsymbol{\theta}) + \begin{bmatrix} \mathbf{C}_{\theta x_1} & \mathbf{C}_{\theta x_2} \end{bmatrix} \begin{bmatrix} \mathbf{C}_{x_1x_1}^{-1} & \mathbf{0} \\ \mathbf{0} & \mathbf{C}_{x_2x_2}^{-1} \end{bmatrix} \begin{bmatrix} \mathbf{x}_1 - E(\mathbf{x}_1) \\ \mathbf{x}_2 - E(\mathbf{x}_2) \end{bmatrix} \\ &= E(\boldsymbol{\theta}) + \mathbf{C}_{\theta x_1}\mathbf{C}_{x_1x_1}^{-1}(\mathbf{x}_1 - E(\mathbf{x}_1)) + \mathbf{C}_{\theta x_2}\mathbf{C}_{x_2x_2}^{-1}(\mathbf{x}_2 - E(\mathbf{x}_2)). \end{aligned}$$

We may interpret the estimator as composed of the prior estimator $E(\boldsymbol{\theta})$ as well as that due to the independent data sets. The MMSE is seen to have an additivity property for *independent* data sets. This result is useful in deriving the sequential MMSE estimator (see Chapter 13).

Finally, in the jointly Gaussian case the MMSE is linear in the data, as can be seen from (11.17). This will allow us to easily determine the PDF of the error, as described in Section 11.6.

11.5 Maximum A Posteriori Estimators

In the MAP estimation approach we choose $\hat{\theta}$ to maximize the posterior PDF or

$$\hat{\theta} = \arg\max_{\theta} p(\theta|\mathbf{x}).$$

This was shown to minimize the Bayes risk for a "hit-or-miss" cost function. In finding the maximum of $p(\theta|\mathbf{x})$ we observe that

$$p(\theta|\mathbf{x}) = \frac{p(\mathbf{x}|\theta)p(\theta)}{p(\mathbf{x})}$$

so an equivalent maximization is of $p(\mathbf{x}|\theta)p(\theta)$. This is reminiscent of the MLE except for the presence of the prior PDF. Hence, the MAP estimator is

$$\hat{\theta} = \arg\max_{\theta} p(\mathbf{x}|\theta)p(\theta) \tag{11.18}$$

or, equivalently,

$$\hat{\theta} = \arg\max_{\theta} [\ln p(\mathbf{x}|\theta) + \ln p(\theta)] \tag{11.19}$$

Before extending the MAP estimator to the vector case we give some examples.

Example 11.2 - Exponential PDF

Assume that

$$p(x[n]|\theta) = \begin{cases} \theta \exp(-\theta x[n]) & x[n] > 0 \\ 0 & x[n] < 0 \end{cases}$$

where the $x[n]$'s are conditionally IID, or

$$p(\mathbf{x}|\theta) = \prod_{n=0}^{N-1} p(x[n]|\theta)$$

and the prior PDF is

$$p(\theta) = \begin{cases} \lambda \exp(-\lambda\theta) & \theta > 0 \\ 0 & \theta < 0. \end{cases}$$

Then, the MAP estimator is found by maximizing

$$\begin{aligned} g(\theta) &= \ln p(\mathbf{x}|\theta) + \ln p(\theta) \\ &= \ln \left[\theta^N \exp\left(-\theta \sum_{n=0}^{N-1} x[n] \right) \right] + \ln [\lambda \exp(-\lambda\theta)] \\ &= N \ln \theta - N\theta\bar{x} + \ln \lambda - \lambda\theta \end{aligned}$$

for $\theta > 0$. Differentiating with respect to θ produces

$$\frac{dg(\theta)}{d\theta} = \frac{N}{\theta} - N\bar{x} - \lambda$$

and setting it equal to zero yields the MAP estimator

$$\hat{\theta} = \frac{1}{\bar{x} + \frac{\lambda}{N}}.$$

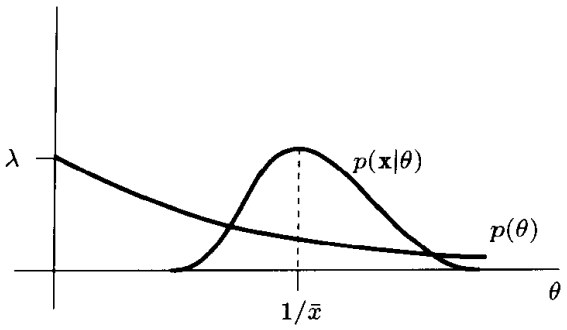


Figure 11.3 Domination of prior PDF by conditional PDF in MAP estimator

Note that as $N \to \infty$, $\hat{\theta} \to 1/\bar{x}$. Also, recall that $E(x[n]|\theta) = 1/\theta$ (see Example 9.2), so that

$$\theta = \frac{1}{E(x[n]|\theta)}$$

confirming the reasonableness of the MAP estimator. Also, if $\lambda \to 0$ so that the prior PDF is nearly uniform, we obtain the estimator $1/\bar{x}$. In fact, this is the *Bayesian MLE* (the estimator obtained by maximizing $p(\mathbf{x}|\theta)$) since as $\lambda \to 0$ we have the situation in Figure 11.3 in which the conditional PDF *dominates* the prior PDF. The maximum of g is then unaffected by the prior PDF. ◇

Example 11.3 - DC Level in WGN - Uniform Prior PDF

Recall the introductory example in Chapter 10. There we discussed the MMSE estimator of A for a DC level in WGN with a uniform prior PDF. The MMSE estimator as given by (10.9) could not be obtained in explicit form due to the need to evaluate the integrals. The posterior PDF was given as

$$p(A|\mathbf{x}) = \begin{cases} \dfrac{\frac{1}{\sqrt{2\pi\sigma^2/N}} \exp\left[-\frac{1}{2\frac{\sigma^2}{N}}(A-\bar{x})^2\right]}{\int_{-A_0}^{A_0} \frac{1}{\sqrt{2\pi\sigma^2/N}} \exp\left[-\frac{1}{2\frac{\sigma^2}{N}}(A-\bar{x})^2\right] dA} & |A| \le A_0 \\ 0 & |A| > A_0. \end{cases}$$

In determining the MAP estimator of A we first note that the denominator or normalizing factor does not depend on A. This observation formed the basis for (11.18). The posterior PDF is shown in Figure 11.4 for various values of $\bar{x}$. The dashed curves indicate the portion of the original Gaussian PDF that was truncated to yield the posterior PDF. It should be evident from the figure that the maximum is $\bar{x}$ if $|\bar{x}| \le A_0$, A_0 if $\bar{x} > A_0$, and $-A_0$ if $\bar{x} < -A_0$. Thus, the MAP estimator is explicitly given as

$$\hat{A} = \begin{cases} -A_0 & \bar{x} < -A_0 \\ \bar{x} & -A_0 \le \bar{x} \le A_0 \\ A_0 & \bar{x} > A_0. \end{cases} \tag{11.20}$$

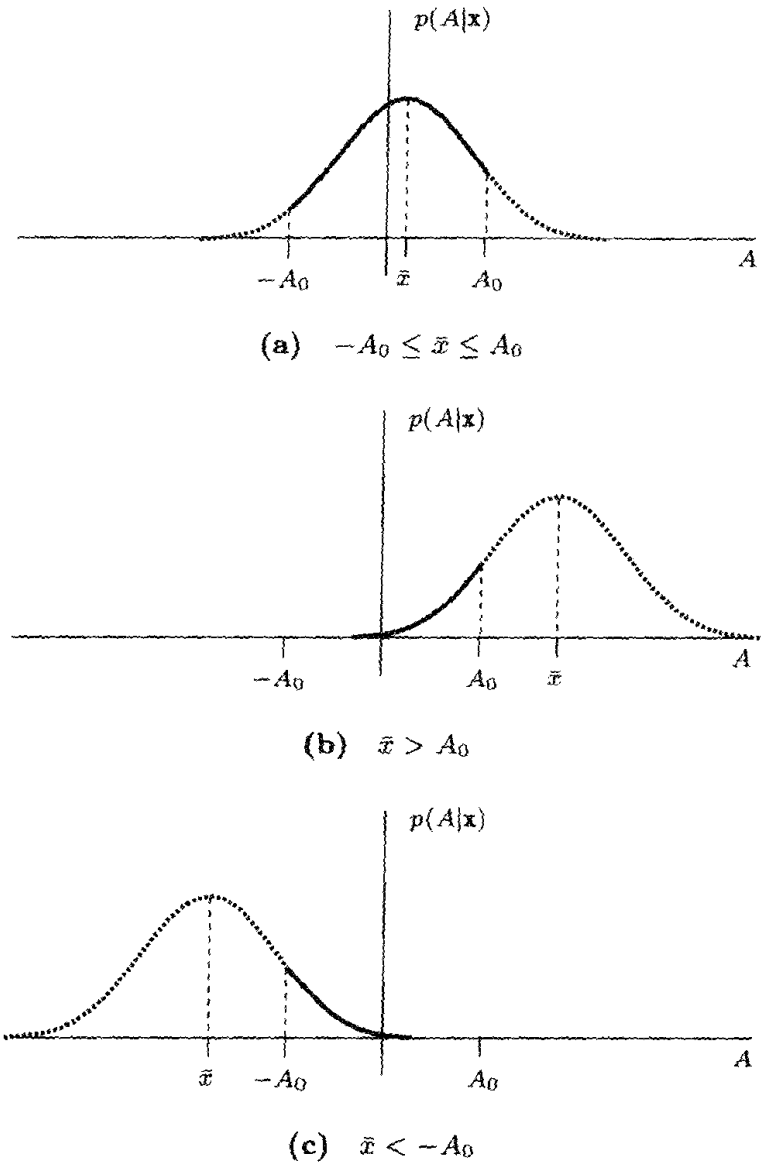


Figure 11.4 Determination of MAP estimator

The "classical estimator" $\bar{x}$ is used unless it falls outside the known constraint interval, in which case we discard the data and rely only on the prior knowledge. Note that the MAP estimator is usually easier to determine since it does not involve any integration but only a maximization. Since $p(\theta)$ is the prior PDF, which we must specify, and $p(\mathbf{x}|\theta)$ is the PDF of the data, which we also assume known in formulating our data model, we see that integration is not required. In the case of the MMSE estimator we saw that integration was a necessary part of the estimator implementation. Of course, the required maximization is not without its potential problems, as evidenced by the difficulties encountered in determining the MLE (see Section 7.7). ◇

To extend the MAP estimator to the vector parameter case in which the posterior PDF is now $p(\boldsymbol{\theta}|\mathbf{x})$, we again employ the result

$$p(\theta_1|\mathbf{x}) = \int \cdots \int p(\boldsymbol{\theta}|\mathbf{x})\, d\theta_2 \ldots d\theta_p. \tag{11.21}$$

Then, the MAP estimator is given by

$$\hat{\theta}_1 = \arg\max_{\theta_1} p(\theta_1|\mathbf{x})$$

or in general

$$\hat{\theta}_i = \arg\max_{\theta_i} p(\theta_i|\mathbf{x}) \qquad i = 1, 2, \ldots, p. \tag{11.22}$$

This estimator minimizes the average "hit-or-miss" cost function

$$\mathcal{R}_i = E[\mathcal{C}(\theta_i - \hat{\theta}_i)]$$

for each i, where the expectation is over $p(\mathbf{x}, \theta_i)$.

One of the advantages of the MAP estimator for a scalar parameter is that to numerically determine it we need only maximize $p(\mathbf{x}|\theta)p(\theta)$. No integration is required. This desirable property of the MAP estimator for a scalar parameter does not carry over to the vector parameter case due to the need to obtain $p(\theta_i|\mathbf{x})$ as per (11.21). However, we might propose the following *vector* MAP estimator

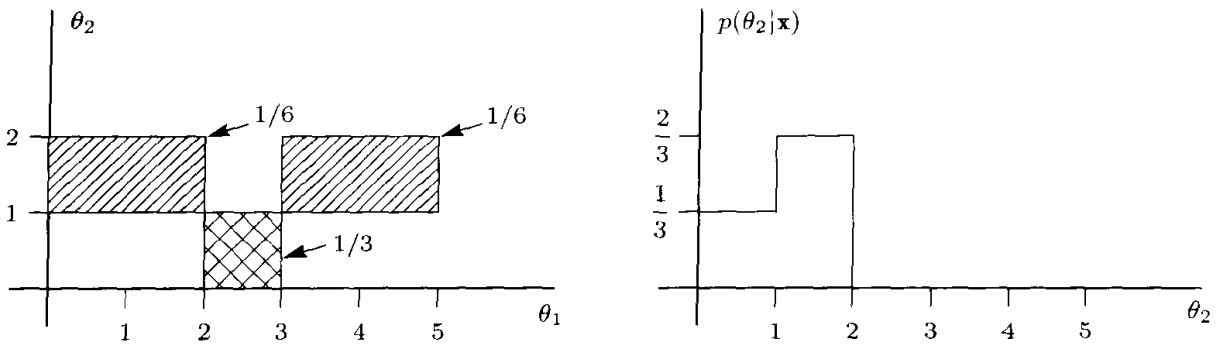
$$\hat{\boldsymbol{\theta}} = \arg\max_{\boldsymbol{\theta}} p(\boldsymbol{\theta}|\mathbf{x}) \tag{11.23}$$

in which the posterior PDF for the vector parameter $\boldsymbol{\theta}$ is maximized to find the estimator. Now we no longer need to determine the marginal PDFs, eliminating the integration steps, since, equivalently,

$$\hat{\boldsymbol{\theta}} = \arg\max_{\boldsymbol{\theta}} p(\mathbf{x}|\boldsymbol{\theta})p(\boldsymbol{\theta}) \tag{11.24}$$

That this estimator is not in general the same as (11.22) is illustrated by the example shown in Figure 11.5. Note that $p(\theta_1, \theta_2|\mathbf{x})$ is constant and equal to $1/6$ on the rectangular regions shown (single cross-hatched) and equal to $1/3$ on the square region (double cross-hatched). It is clear from (11.23) that the *vector* MAP estimator, although not unique, is any value of $\boldsymbol{\theta}$ within the square so that $0 < \hat{\theta}_2 < 1$. The MAP estimator of θ_2 is found as the value maximizing $p(\theta_2|\mathbf{x})$. But

$$\begin{aligned} p(\theta_2|\mathbf{x}) &= \int p(\theta_1, \theta_2|\mathbf{x})\, d\theta_1 \\ &= \begin{cases} \displaystyle\int_2^3 \frac{1}{3}\, d\theta_1 & 0 < \theta_2 < 1 \\ \displaystyle\int_0^2 \frac{1}{6}\, d\theta_1 + \int_3^5 \frac{1}{6}\, d\theta_1 & 1 < \theta_2 < 2 \end{cases} \\ &= \begin{cases} \frac{1}{3} & 0 < \theta_2 < 1 \\ \frac{2}{3} & 1 < \theta_2 < 2 \end{cases} \end{aligned}$$



(a) Posterior PDF $p(\theta_1, \theta_2|\mathbf{x})$ **(b)** Posterior PDF $p(\theta_2|\mathbf{x})$

Figure 11.5 Comparison of scalar and vector MAP estimators

and is shown in Figure 11.5b. Clearly, the MAP estimator is any value $1 < \hat{\theta}_2 < 2$, which differs from the vector MAP estimator. It can be shown, however, that the vector MAP estimator does indeed minimize a Bayes risk, although a different one than the "hit-or-miss" cost function (see Problem 11.11). We will henceforth refer to the vector MAP estimator as just the MAP estimator.

Example 11.4 - DC Level in WGN - Unknown Variance

We observe

$$x[n] = A + w[n] \qquad n = 0, 1, \ldots, N-1.$$

In contrast to the usual example in which only A is to be estimated, we now assume the variance σ^2 of the WGN $w[n]$ is also unknown. The vector parameter is $\boldsymbol{\theta} = [A\ \sigma^2]^T$. We assume the conditional PDF

$$p(\mathbf{x}|A, \sigma^2) = \frac{1}{(2\pi\sigma^2)^{\frac{N}{2}}} \exp\left[-\frac{1}{2\sigma^2}\sum_{n=0}^{N-1}(x[n] - A)^2\right]$$

for the data, so that conditioned on knowing $\boldsymbol{\theta}$, we have a DC level in WGN. The prior PDF is assumed to be

$$\begin{aligned} p(A, \sigma^2) &= p(A|\sigma^2)p(\sigma^2) \\ &= \frac{1}{\sqrt{2\pi\alpha\sigma^2}} \exp\left[-\frac{1}{2\alpha\sigma^2}(A - \mu_A)^2\right] \frac{\lambda \exp(-\frac{\lambda}{\sigma^2})}{\sigma^4}. \end{aligned}$$

The prior PDF of A conditioned on σ^2 is the usual $\mathcal{N}(\mu_A, \sigma_A^2)$ PDF except that $\sigma_A^2 = \alpha\sigma^2$. The prior PDF for σ^2 is the same one used in Example 10.3 and is a special case of the *inverted* gamma PDF. Note that A and σ^2 are not independent. This prior PDF is the conjugate prior PDF (see Problem 10.10). To find the MAP estimator we need to maximize

$$\begin{aligned} g(A, \sigma^2) &= p(\mathbf{x}|A, \sigma^2)p(A, \sigma^2) \\ &= p(\mathbf{x}|A, \sigma^2)p(A|\sigma^2)p(\sigma^2) \end{aligned}$$

over A and σ^2. To find the value of A that maximizes g we can equivalently maximize just $p(\mathbf{x}|A,\sigma^2)p(A|\sigma^2)$. But in Example 10.1 we showed that

$$\begin{aligned} h(A) &= p(\mathbf{x}|A,\sigma^2)p(A|\sigma^2) \\ &= \frac{1}{(2\pi\sigma^2)^{\frac{N}{2}}\sqrt{2\pi\sigma_A^2}} \exp\left[-\frac{1}{2\sigma^2}\sum_{n=0}^{N-1}x^2[n]\right]\exp\left[-\frac{1}{2}Q(A)\right] \end{aligned}$$

where

$$Q(A) = \frac{1}{\sigma_{A|x}^2}(A-\mu_{A|x})^2 - \frac{\mu_{A|x}^2}{\sigma_{A|x}^2} + \frac{\mu_A^2}{\sigma_A^2}$$

and

$$\begin{aligned} \mu_{A|x} &= \frac{\frac{N}{\sigma^2}\bar{x} + \frac{\mu_A}{\sigma_A^2}}{\frac{N}{\sigma^2} + \frac{1}{\sigma_A^2}} \\ \sigma_{A|x}^2 &= \frac{1}{\frac{N}{\sigma^2} + \frac{1}{\sigma_A^2}}. \end{aligned}$$

Maximizing $h(A)$ over A is equivalent to minimizing $Q(A)$, so that

$$\hat{A} = \mu_{A|x} = \frac{\frac{N}{\sigma^2}\bar{x} + \frac{\mu_A}{\sigma_A^2}}{\frac{N}{\sigma^2} + \frac{1}{\sigma_A^2}}$$

and since $\sigma_A^2 = \alpha\sigma^2$, we have

$$\hat{A} = \frac{N\bar{x} + \frac{\mu_A}{\alpha}}{N + \frac{1}{\alpha}}.$$

Since this does not depend on σ^2, it is the MAP estimator. We next find the MAP estimator for σ^2 by maximizing $g(\hat{A},\sigma^2) = h(\hat{A})p(\sigma^2)$. First, note that

$$h(\hat{A}) = \frac{1}{(2\pi\sigma^2)^{\frac{N}{2}}\sqrt{2\pi\sigma_A^2}} \exp\left[-\frac{1}{2\sigma^2}\sum_{n=0}^{N-1}x^2[n]\right]\exp\left[-\frac{1}{2}Q(\hat{A})\right]$$

where

$$Q(\hat{A}) = \frac{\mu_A^2}{\sigma_A^2} - \frac{\mu_{A|x}^2}{\sigma_{A|x}^2}.$$

Letting $\sigma_A^2 = \alpha\sigma^2$, we have

$$\begin{aligned} Q(\hat{A}) &= \frac{\mu_A^2}{\alpha\sigma^2} - \hat{A}^2\left(\frac{N}{\sigma^2} + \frac{1}{\alpha\sigma^2}\right) \\ &= \frac{1}{\sigma^2}\left[\underbrace{\frac{\mu_A^2}{\alpha} - \hat{A}^2(N+\frac{1}{\alpha})}_{\xi}\right]. \end{aligned}$$

Therefore, we must maximize over σ^2

$$\begin{aligned} g(\hat{A},\sigma^2) &= \frac{1}{(2\pi\sigma^2)^{\frac{N}{2}}\sqrt{2\pi\alpha\sigma^2}} \exp\left[-\frac{1}{2\sigma^2}\sum_{n=0}^{N-1}x^2[n]-\frac{1}{2\sigma^2}\xi\right]\frac{\lambda\exp(-\frac{\lambda}{\sigma^2})}{\sigma^4} \\ &= \frac{c}{(\sigma^2)^{\frac{N+5}{2}}}\exp\left(-\frac{a}{\sigma^2}\right) \end{aligned}$$

where c is a constant independent of σ^2 and

$$a = \frac{1}{2}\sum_{n=0}^{N-1}x^2[n]+\frac{1}{2}\xi+\lambda.$$

Differentiating $\ln g$ produces

$$\frac{\partial \ln g(\hat{A},\sigma^2)}{\partial\sigma^2} = -\frac{\frac{N+5}{2}}{\sigma^2}+\frac{a}{\sigma^4}$$

and setting it to zero yields

$$\begin{aligned} \hat{\sigma^2} &= \frac{1}{\frac{N+5}{2}}a \\ &= \frac{1}{\frac{N+5}{2}}\left[\frac{1}{2}\sum_{n=0}^{N-1}x^2[n]+\frac{1}{2}\xi+\lambda\right] \\ &= \frac{1}{N+5}\left[\sum_{n=0}^{N-1}x^2[n]+\frac{\mu_A^2}{\alpha}-\hat{A}^2\left(N+\frac{1}{\alpha}\right)+2\lambda\right] \\ &= \frac{N}{N+5}\left[\frac{1}{N}\sum_{n=0}^{N-1}x^2[n]-\hat{A}^2\right]+\frac{1}{(N+5)\alpha}(\mu_A^2-\hat{A}^2)+\frac{2\lambda}{N+5}. \end{aligned}$$

The MAP estimator is therefore

$$\begin{aligned} \hat{\boldsymbol{\theta}} &= \begin{bmatrix} \hat{A} \\ \hat{\sigma^2} \end{bmatrix} \\ &= \begin{bmatrix} \frac{N\bar{x}+\frac{\mu_A}{\alpha}}{N+\frac{1}{\alpha}} \\ \frac{N}{N+5}\left[\frac{1}{N}\sum_{n=0}^{N-1}x^2[n]-\hat{A}^2\right]+\frac{1}{(N+5)\alpha}(\mu_A^2-\hat{A}^2)+\frac{2\lambda}{N+5} \end{bmatrix}. \end{aligned}$$

As expected, if $N\to\infty$ so that the conditional or data PDF dominates the prior PDF, we have

$$\begin{aligned} \hat{A} &\to \bar{x} \\ \hat{\sigma^2} &\to \frac{1}{N}\sum_{n=0}^{N-1}x^2[n]-\bar{x}^2 = \frac{1}{N}\sum_{n=0}^{N-1}(x[n]-\bar{x})^2 \end{aligned}$$

which is the Bayesian MLE. Recall that the Bayesian MLE is the value of $\boldsymbol{\theta}$ that maximizes $p(\mathbf{x}|\boldsymbol{\theta})$. $\diamond$

It is true in general that as $N \to \infty$, the MAP estimator becomes the Bayesian MLE. To find the MAP estimator we must maximize $p(\mathbf{x}|\boldsymbol{\theta})p(\boldsymbol{\theta})$. If the prior $p(\boldsymbol{\theta})$ is uniform over the range of $\boldsymbol{\theta}$ for which $p(\mathbf{x}|\boldsymbol{\theta})$ is essentially nonzero (as will be the case if $N \to \infty$ so that the data PDF dominates the prior PDF), then the maximization will be equivalent to a maximization of $p(\mathbf{x}|\boldsymbol{\theta})$. If $p(\mathbf{x}|\boldsymbol{\theta})$ has the same form as the PDF family $p(\mathbf{x};\boldsymbol{\theta})$ (as it does for the previous example), then the Bayesian MLE and the classical MLE will have the same form. The reader is cautioned, however, that the estimators are inherently different due to the contrasting underlying experiments.

Some properties of the MAP estimator are of importance. First, if the posterior PDF is Gaussian, as in the Bayesian linear model for example, the mode or peak location is identical to the mean. Hence, *the MAP estimator is identical to the MMSE estimator if* $\mathbf{x}$ *and* $\boldsymbol{\theta}$ *are jointly Gaussian.* Second, the invariance property encountered in maximum likelihood theory *does not hold* for the MAP estimator. The next example illustrates this.

Example 11.5 - Exponential PDF

Assume in Example 11.2 that we wish to estimate $\alpha = 1/\theta$. It might be supposed that the MAP estimator is

$$\hat{\alpha} = \frac{1}{\hat{\theta}}$$

where $\hat{\theta}$ is the MAP estimator of θ, so that

$$\hat{\alpha} = \bar{x} + \frac{\lambda}{N}. \tag{11.25}$$

We now show that *this is not true.* As before

$$p(x[n]|\theta) = \begin{cases} \theta \exp(-\theta x[n]) & x[n] > 0 \\ 0 & x[n] < 0. \end{cases}$$

The conditional PDF based on observing α is

$$p(x[n]|\alpha) = \begin{cases} \frac{1}{\alpha} \exp\left(-\frac{x[n]}{\alpha}\right) & x[n] > 0 \\ 0 & x[n] < 0 \end{cases}$$

since knowing α is equivalent to knowing θ. The prior PDF

$$p(\theta) = \begin{cases} \lambda \exp(-\lambda\theta) & \theta > 0 \\ 0 & \theta < 0 \end{cases}$$

cannot be transformed to $p(\alpha)$ by just letting $\theta = 1/\alpha$. This is because θ is a *random variable*, not a deterministic parameter. The PDF of α must account for the derivative

of the transformation or

$$\begin{aligned} p_\alpha(\alpha) &= \frac{p_\theta(\theta(\alpha))}{\left|\frac{d\alpha}{d\theta}\right|} \\ &= \begin{cases} \frac{\lambda \exp(-\lambda/\alpha)}{\alpha^2} & \alpha > 0 \\ 0 & \alpha < 0. \end{cases} \end{aligned}$$

Otherwise, the prior PDF would not integrate to 1. Completing the problem

$$\begin{aligned} g(\alpha) &= \ln p(\mathbf{x}|\alpha) + \ln p(\alpha) \\ &= \ln \left[\left(\frac{1}{\alpha}\right)^N \exp\left(-\frac{1}{\alpha}\sum_{n=0}^{N-1} x[n]\right)\right] + \ln \frac{\lambda \exp(-\lambda/\alpha)}{\alpha^2} \\ &= -N\ln\alpha - N\frac{\bar{x}}{\alpha} + \ln\lambda - \frac{\lambda}{\alpha} - 2\ln\alpha \\ &= -(N+2)\ln\alpha - \frac{N\bar{x}+\lambda}{\alpha} + \ln\lambda. \end{aligned}$$

Differentiating with respect to α

$$\frac{dg}{d\alpha} = -\frac{N+2}{\alpha} + \frac{N\bar{x}+\lambda}{\alpha^2}$$

and setting it equal to zero yields the MAP estimator

$$\hat{\alpha} = \frac{N\bar{x}+\lambda}{N+2}$$

which is not the same as (11.25). Hence, it is seen that that MAP estimator does not commute over nonlinear transformations, although it does so for linear transformations (see Problem 11.12).

◇

11.6 Performance Description

In the classical estimation problem we were interested in the mean and variance of an estimator. Assuming the estimator was Gaussian, we could then immediately obtain the PDF. If the PDF was concentrated about the true value of the parameter, then we could say that the estimator performed well. In the case of a random parameter the same approach cannot be used. Now the randomness of the parameter results in a different PDF of the estimator for each *realization* of θ. We denote this PDF by $p(\hat{\theta}|\theta)$. To perform well the estimate should be close to θ for every possible value of θ or the error

$$\epsilon = \theta - \hat{\theta}$$

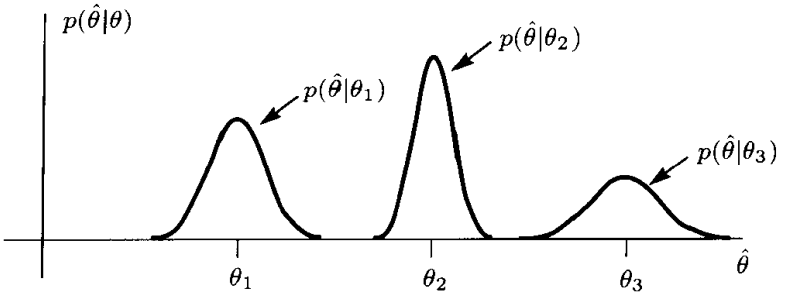


Figure 11.6 Dependence of estimation error on realization of parameter

should be small. This is illustrated in Figure 11.6 where several conditional PDFs are shown. Each PDF corresponds to that obtained for a given realization of θ. For $\hat{\theta}$ to be a good estimator $p(\hat{\theta}|\theta)$ should be concentrated about θ for all possible θ. This reasoning led to the MMSE estimator, which minimizes $E_{x,\theta}[(\theta-\hat{\theta})^2]$. For an arbitrary Bayesian estimator it thus makes sense to assess the performance by determining *the PDF of the error*. This PDF, which now accounts for the randomness of θ, should be concentrated about zero. For the MMSE estimator $\hat{\theta} = E(\theta|\mathbf{x})$, the error is

$$\epsilon = \theta - E(\theta|\mathbf{x})$$

and the mean of the error is

$$\begin{aligned} E_{x,\theta}(\epsilon) &= E_{x,\theta}\left[\theta - E(\theta|\mathbf{x})\right] \\ &= E_x\left[E_{\theta|x}(\theta) - E_{\theta|x}(\theta|\mathbf{x})\right] \\ &= E_x\left[E(\theta|\mathbf{x}) - E(\theta|\mathbf{x})\right] = 0 \end{aligned}$$

so that the error of the MMSE estimator is on the average (with respect to $p(\mathbf{x},\theta)$) zero. The variance of the error for the MMSE estimator is

$$\text{var}(\epsilon) = E_{x,\theta}(\epsilon^2)$$

since $E(\epsilon) = 0$, and therefore

$$\text{var}(\epsilon) = E_{x,\theta}\left[(\theta-\hat{\theta})^2\right]$$

which is just the minimum Bayesian MSE. Finally, if ϵ is Gaussian, we have that

$$\epsilon \sim \mathcal{N}(0, \text{Bmse}(\hat{\theta})). \tag{11.26}$$

Example 11.6 - DC Level in WGN - Gaussian Prior PDF

Referring to Example 10.1, we have that

$$\hat{A} = \frac{\sigma_A^2}{\sigma_A^2 + \frac{\sigma^2}{N}}\bar{x} + \frac{\frac{\sigma^2}{N}}{\sigma_A^2 + \frac{\sigma^2}{N}}\mu_A$$

and

$$\text{Bmse}(\hat{A}) = \frac{1}{\frac{N}{\sigma^2} + \frac{1}{\sigma_A^2}}.$$

In this case the error is $\epsilon = A - \hat{A}$. Because $\hat{A}$ depends linearly on $\mathbf{x}$, and $\mathbf{x}$ and A are jointly Gaussian, ϵ is Gaussian. As a result, we have from (11.26)

$$\epsilon \sim \mathcal{N}\left(0, \frac{1}{\frac{N}{\sigma^2} + \frac{1}{\sigma_A^2}}\right).$$

As $N \to \infty$, the PDF collapses about zero and the estimator can be said to be *consistent in the Bayesian sense.* Consistency means that for a large enough data record $\hat{A}$ will always be close to the *realization* of A, regardless of the realization value (see also Problem 11.10). ◇

For a vector parameter $\boldsymbol{\theta}$ the error can be defined as

$$\boldsymbol{\epsilon} = \boldsymbol{\theta} - \hat{\boldsymbol{\theta}}.$$

As before, if $\hat{\boldsymbol{\theta}}$ is the MMSE estimator, then $\boldsymbol{\epsilon}$ will have a zero mean. Its covariance matrix is

$$E_{x,\theta}(\boldsymbol{\epsilon}\boldsymbol{\epsilon}^T) = E_{x,\theta}\left[\left(\boldsymbol{\theta} - E(\boldsymbol{\theta}|\mathbf{x})\right)\left(\boldsymbol{\theta} - E(\boldsymbol{\theta}|\mathbf{x})\right)^T\right].$$

The expectation is with respect to the PDF $p(\mathbf{x}, \boldsymbol{\theta})$. Note that

$$\begin{aligned}[E(\boldsymbol{\theta}|\mathbf{x})]_i = E(\theta_i|\mathbf{x}) &= \int \theta_i p(\boldsymbol{\theta}|\mathbf{x})\, d\boldsymbol{\theta} \\ &= \int \theta_i p(\theta_i|\mathbf{x})\, d\theta_i\end{aligned}$$

and depends on $\mathbf{x}$ only. Hence, $E(\boldsymbol{\theta}|\mathbf{x})$ is a function of $\mathbf{x}$, as it should be for a valid estimator. As in the scalar parameter case, the diagonal elements of the covariance matrix of the error represent the minimum Bayesian MSE since the $[i, i]$ element is

$$\left[E_{x,\theta}(\boldsymbol{\epsilon}\boldsymbol{\epsilon}^T)\right]_{ii} = \int\int \left[\theta_i - E(\theta_i|\mathbf{x})\right]^2 p(\mathbf{x}, \boldsymbol{\theta})\, d\mathbf{x}\, d\boldsymbol{\theta}.$$

Integrating with respect to $\theta_1, \ldots, \theta_{i-1}, \theta_{i+1}, \ldots, \theta_p$ produces

$$\left[E_{x,\theta}(\boldsymbol{\epsilon}\boldsymbol{\epsilon}^T)\right]_{ii} = \int\int \left[\theta_i - E(\theta_i|\mathbf{x})\right]^2 p(\mathbf{x}, \theta_i)\, d\mathbf{x}\, d\theta_i$$

which is the minimum Bayesian MSE for $\hat{\theta}_i$. Hence, the diagonal elements of the error covariance matrix $E_{x,\theta}(\boldsymbol{\epsilon}\boldsymbol{\epsilon}^T)$ are just the minimum Bayesian MSEs. The error covariance matrix is sometimes called the *Bayesian mean square error matrix* and is

termed $\mathbf{M}_{\hat{\theta}}$. To determine the PDF of $\boldsymbol{\epsilon}$ we now derive the entire covariance matrix. We specialize our discussion to the Bayesian linear model (see Theorem 10.3). Then,

$$\begin{aligned}\mathbf{M}_{\hat{\theta}} &= E_{x,\theta}\left[(\boldsymbol{\theta} - E(\boldsymbol{\theta}|\mathbf{x}))(\boldsymbol{\theta} - E(\boldsymbol{\theta}|\mathbf{x}))^T\right] \\ &= E_x E_{\theta|x}\left[(\boldsymbol{\theta} - E(\boldsymbol{\theta}|\mathbf{x}))(\boldsymbol{\theta} - E(\boldsymbol{\theta}|\mathbf{x}))^T\right] \\ &= E_x(\mathbf{C}_{\theta|x})\end{aligned}$$

where $\mathbf{C}_{\theta|x}$ is the covariance matrix of the posterior PDF $p(\boldsymbol{\theta}|\mathbf{x})$. If $\boldsymbol{\theta}$ and $\mathbf{x}$ are jointly Gaussian, then from (10.25)

$$\mathbf{M}_{\hat{\theta}} = \mathbf{C}_{\theta\theta} - \mathbf{C}_{\theta x}\mathbf{C}_{xx}^{-1}\mathbf{C}_{x\theta} \tag{11.27}$$

since $\mathbf{C}_{\theta|x}$ does not depend on $\mathbf{x}$. Specializing to the Bayesian linear model, we have from (10.29) (shortening the notation from $\mathbf{C}_{\theta\theta}$ to $\mathbf{C}_\theta$)

$$\mathbf{M}_{\hat{\theta}} = \mathbf{C}_\theta - \mathbf{C}_\theta\mathbf{H}^T(\mathbf{H}\mathbf{C}_\theta\mathbf{H}^T + \mathbf{C}_w)^{-1}\mathbf{H}\mathbf{C}_\theta \tag{11.28}$$

or from (10.33)

$$\mathbf{M}_{\hat{\theta}} = \left(\mathbf{C}_\theta^{-1} + \mathbf{H}^T\mathbf{C}_w^{-1}\mathbf{H}\right)^{-1}. \tag{11.29}$$

Finally, it should be observed that for the Bayesian linear model $\boldsymbol{\epsilon} = \boldsymbol{\theta} - \hat{\boldsymbol{\theta}}$ is Gaussian since from (10.28)

$$\begin{aligned}\boldsymbol{\epsilon} &= \boldsymbol{\theta} - \hat{\boldsymbol{\theta}} \\ &= \boldsymbol{\theta} - \boldsymbol{\mu}_\theta - \mathbf{C}_\theta\mathbf{H}^T(\mathbf{H}\mathbf{C}_\theta\mathbf{H}^T + \mathbf{C}_w)^{-1}(\mathbf{x} - \mathbf{H}\boldsymbol{\mu}_\theta)\end{aligned}$$

and thus $\boldsymbol{\epsilon}$ is a linear transformation of $\mathbf{x}, \boldsymbol{\theta}$, which are themselves jointly Gaussian. Hence, the error vector for the MMSE estimator of the parameters of a Bayesian linear model is characterized by

$$\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{M}_{\hat{\theta}}) \tag{11.30}$$

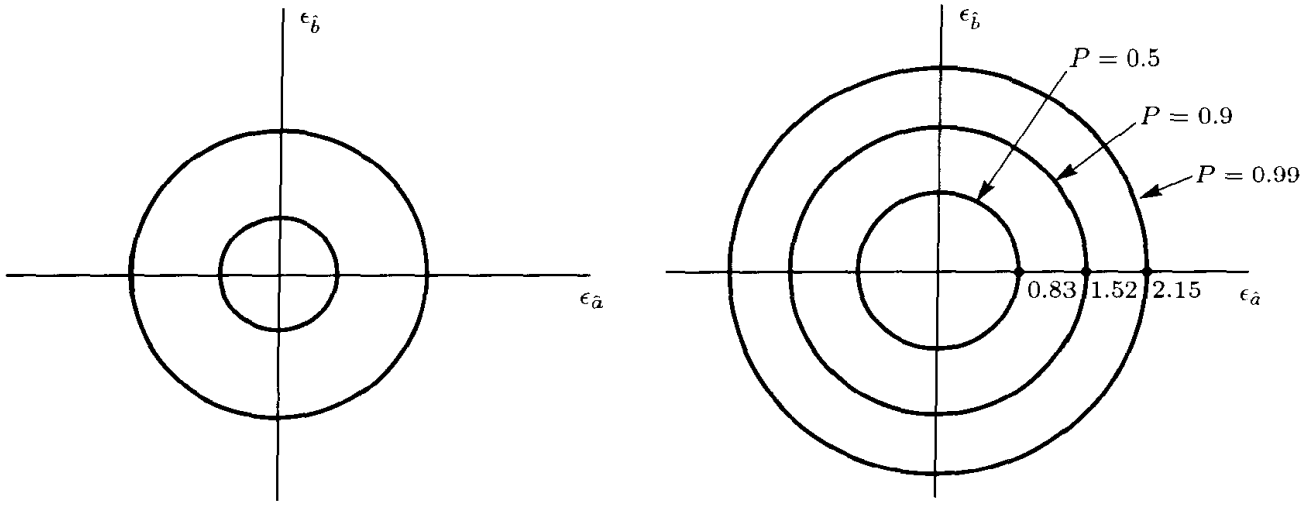
where $\mathbf{M}_{\hat{\theta}}$ is given by (11.28) or by (11.29). An example follows.

Example 11.7 - Bayesian Fourier Analysis (continued)

We now again consider Example 11.1. Recall that

$$\begin{aligned}\mathbf{C}_\theta &= \sigma_\theta^2\mathbf{I} \\ \mathbf{C}_w &= \sigma^2\mathbf{I} \\ \mathbf{H}^T\mathbf{H} &= \frac{N}{2}\mathbf{I}.\end{aligned}$$

Hence, from (11.29)



(a) Contours of constant probability density **(b)** Error "ellipses" for $\sigma_\theta^2 = 1$, $\sigma^2/N = 1/2$

Figure 11.7 Error ellipses for Bayesian Fourier analysis

$$\begin{aligned}\mathbf{M}_{\hat{\theta}} &= \left(\mathbf{C}_\theta^{-1} + \mathbf{H}^T\mathbf{C}_w^{-1}\mathbf{H}\right)^{-1} \\ &= \left(\frac{1}{\sigma_\theta^2}\mathbf{I} + \frac{N}{2\sigma^2}\mathbf{I}\right)^{-1} \\ &= \left(\frac{1}{\sigma_\theta^2} + \frac{N}{2\sigma^2}\right)^{-1}\mathbf{I}.\end{aligned}$$

Therefore, the error $\boldsymbol{\epsilon} = [\epsilon_{\hat{a}}\ \epsilon_{\hat{b}}]^T$ has the PDF

$$\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{M}_{\hat{\theta}})$$

where

$$\mathbf{M}_{\hat{\theta}} = \begin{bmatrix} \frac{1}{\frac{1}{\sigma_\theta^2} + \frac{1}{2\sigma^2/N}} & 0 \\ 0 & \frac{1}{\frac{1}{\sigma_\theta^2} + \frac{1}{2\sigma^2/N}} \end{bmatrix}$$

The error components are seen to be independent and, furthermore,

$$\begin{aligned}\text{Bmse}(\hat{a}) &= [\mathbf{M}_{\hat{\theta}}]_{11} \\ \text{Bmse}(\hat{b}) &= [\mathbf{M}_{\hat{\theta}}]_{22}\end{aligned}$$

in agreement with the results of Example 11.1. Also, for this example the PDF of the error vector has contours of constant probability density that are circular, as shown in Figure 11.7a. One way to summarize the performance of the estimator is by the error

or concentration ellipse (in this case it is a circle). It is in general an ellipse within which the error vector will lie with probability P. Let

$$\boldsymbol{\epsilon}^T \mathbf{M}_{\hat{\theta}}^{-1} \boldsymbol{\epsilon} = c^2. \tag{11.31}$$

Then, the probability P that $\boldsymbol{\epsilon}$ will lie within the ellipse described by (11.31) is

$$P = \Pr\{\boldsymbol{\epsilon}^T \mathbf{M}_{\hat{\theta}}^{-1} \boldsymbol{\epsilon} \leq c^2\}.$$

But $u = \boldsymbol{\epsilon}^T \mathbf{M}_{\hat{\theta}}{}^{-1} \boldsymbol{\epsilon}$ is a χ_2^2 random variable with PDF (see Problem 11.13)

$$p(u) = \begin{cases} \frac{1}{2} \exp(-\frac{u}{2}) & u > 0 \\ 0 & u < 0 \end{cases}$$

so that

$$\begin{aligned} P &= \Pr\{\boldsymbol{\epsilon}^T \mathbf{M}_{\hat{\theta}}^{-1} \boldsymbol{\epsilon} \leq c^2\} \\ &= \int_0^{c^2} \frac{1}{2} \exp\left(-\frac{u}{2}\right) du \\ &= 1 - \exp\left(-\frac{c^2}{2}\right) \end{aligned}$$

or

$$\boldsymbol{\epsilon}^T \mathbf{M}_{\hat{\theta}}^{-1} \boldsymbol{\epsilon} = 2 \ln\left(\frac{1}{1-P}\right)$$

describes the error ellipse for a probability P. The error vector will lie within this ellipse with probability P. An example is shown in Figure 11.7b for $\sigma_\theta^2 = 1, \sigma^2/N = 1/2$ so that

$$\mathbf{M}_{\hat{\theta}} = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{bmatrix}$$

and thus

$$\boldsymbol{\epsilon}^T \boldsymbol{\epsilon} = \ln\left(\frac{1}{1-P}\right).$$

In general, the contours will be elliptical if the minimum MSE matrix is not a scaled identity matrix (see also Problems 11.14 and 11.15). ◇

We now summarize our results in a theorem.

Theorem 11.1 (Performance of the MMSE Estimator for the Bayesian Linear Model) *If the observed data* $\mathbf{x}$ *can be modeled by the Bayesian linear model of Theorem 10.3, the MMSE estimator is*

$$\hat{\boldsymbol{\theta}} = \boldsymbol{\mu}_\theta + \mathbf{C}_\theta \mathbf{H}^T \left(\mathbf{H}\mathbf{C}_\theta \mathbf{H}^T + \mathbf{C}_w\right)^{-1} (\mathbf{x} - \mathbf{H}\boldsymbol{\mu}_\theta) \tag{11.32}$$

$$= \boldsymbol{\mu}_\theta + \left(\mathbf{C}_\theta^{-1} + \mathbf{H}^T \mathbf{C}_w^{-1} \mathbf{H}\right)^{-1} \mathbf{H}^T \mathbf{C}_w^{-1} (\mathbf{x} - \mathbf{H}\boldsymbol{\mu}_\theta). \tag{11.33}$$

The performance of the estimator is measured by the error $\boldsymbol{\epsilon} = \boldsymbol{\theta} - \hat{\boldsymbol{\theta}}$, *whose PDF is Gaussian with mean zero and covariance matrix*

$$\begin{aligned} \mathbf{C}_\epsilon &= E_{x,\theta}(\boldsymbol{\epsilon}\boldsymbol{\epsilon}^T) \\ &= \mathbf{C}_\theta - \mathbf{C}_\theta \mathbf{H}^T \left(\mathbf{H}\mathbf{C}_\theta \mathbf{H}^T + \mathbf{C}_w\right)^{-1} \mathbf{H}\mathbf{C}_\theta \qquad (11.34) \\ &= \left(\mathbf{C}_\theta^{-1} + \mathbf{H}^T \mathbf{C}_w^{-1} \mathbf{H}\right)^{-1}. \qquad (11.35) \end{aligned}$$

The error covariance matrix is also the minimum MSE matrix $\mathbf{M}_{\hat{\theta}}$, *whose diagonal elements yield the minimum Bayesian MSE or*

$$\begin{aligned} [\mathbf{M}_{\hat{\theta}}]_{ii} &= [\mathbf{C}_\epsilon]_{ii} \\ &= \text{Bmse}(\hat{\theta}_i). \qquad (11.36) \end{aligned}$$

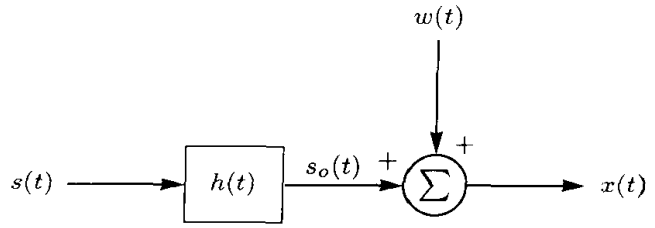
See Problems 11.16 and 11.17 for an application to line fitting.

11.7 Signal Processing Example

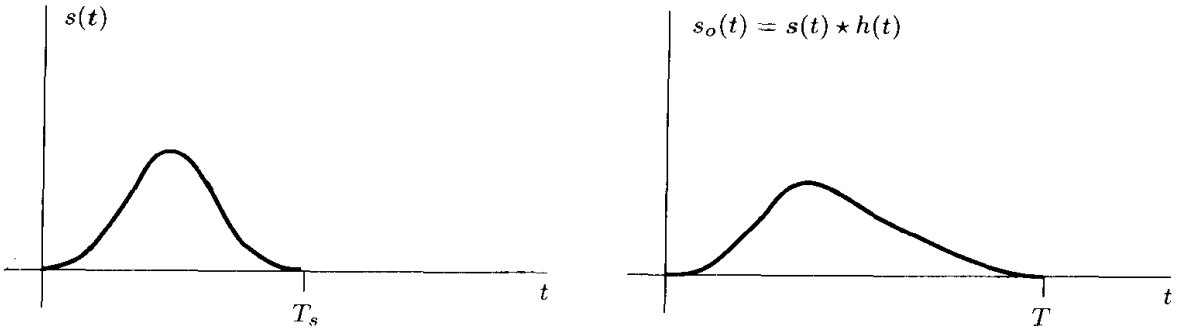
There are many signal processing problems that fit the Bayesian linear model. We mention but a few. Consider a communication problem in which we transmit a signal $s(t)$ through a channel with impulse response $h(t)$. At the channel output we observe a noise corrupted waveform as shown in Figure 11.8. The problem is to estimate $s(t)$ over the interval $0 \leq t \leq T_s$. Because the channel will distort and lengthen the signal, we observe $x(t)$ over the longer interval $0 \leq t \leq T$. Such a problem is sometimes referred to as a *deconvolution* problem [Haykin 1991]. We wish to deconvolve $s(t)$ from a noise corrupted version of $s_o(t) = s(t) \star h(t)$, where $\star$ denotes convolution. This problem also arises in seismic processing in which $s(t)$ represents a series of acoustic reflections of an explosively generated signal due to rock inhomogeneities [Robinson and Treitel 1980]. The filter impulse response $h(t)$ models the earth medium. In image processing the same problem arises for two-dimensional signals in which the two-dimensional version of $s(t)$ represents an image, while the two-dimensional version of $h(t)$ models a distortion due to a poor lens, for example [Jain 1989]. To make any progress in this problem we assume that $h(t)$ is *known*. Otherwise, the problem is one of *blind deconvolution*, which is much more difficult [Haykin 1991]. We further assume that $s(t)$ is a *realization of a random process*. This modeling is appropriate for speech, as an example, for which the signal changes with speaker and content. Hence, the Bayesian assumption appears to be a reasonable one. The observed continuous-time data are

$$x(t) = \int_0^{T_s} h(t-\tau)s(\tau)\,d\tau + w(t) \qquad 0 \leq t \leq T. \qquad (11.37)$$

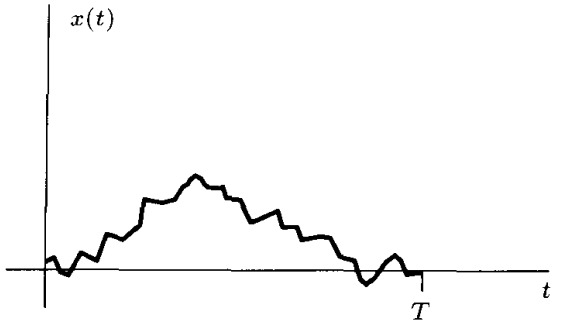
It is assumed that $s(t)$ is nonzero over the interval $[0, T_s]$, and $h(t)$ is nonzero over the interval $[0, T_h]$, so that the observation interval is chosen to be $[0, T]$, where $T = T_s + T_h$. We observe the output signal $s_o(t)$ embedded in noise $w(t)$. In converting the problem



(a) System model



(b) Typical signals



(c) Typical data waveform

Figure 11.8 Generic deconvolution problem

to discrete time we will assume that $s(t)$ is essentially bandlimited to B Hz so that the output signal $s_o(t)$ is also bandlimited to B Hz. The continuous-time noise is assumed to be a zero mean WSS Gaussian random process with PSD

$$P_{ww}(F) = \begin{cases} \frac{N_0}{2} & |F| < B \\ 0 & |F| > B. \end{cases}$$

We sample $x(t)$ at times $t = n\Delta$, where $\Delta = 1/(2B)$, to produce the equivalent discrete-time data (see Appendix 11A)

$$x[n] = \sum_{m=0}^{n_s-1} h[n-m]s[m] + w[n] \qquad n = 0, 1, \ldots, N-1$$

where $x[n] = x(n\Delta), h[n] = \Delta h(n\Delta), s[n] = s(n\Delta)$, and $w[n] = w(n\Delta)$. The discrete-time signal $s[n]$ is nonzero over the interval $[0, n_s - 1]$, and $h[n]$ is nonzero over the interval $[0, n_h - 1]$. The largest integers less than or equal to T_s/Δ and T_h/Δ are denoted by $n_s - 1$ and $n_h - 1$, respectively. Therefore, $N = n_s + n_h - 1$. Finally, it follows that $w[n]$ is WGN with variance $\sigma^2 = N_0 B$. This is because the ACF of $w(t)$ is a sinc function and the samples of $w(t)$ correspond to the zeros of the sinc function (see Example 3.13). If we assume that $s(t)$ is a Gaussian process, then $s[n]$ is a discrete-time Gaussian process. In vector-matrix notation we have

$$\underbrace{\begin{bmatrix} x[0] \\ x[1] \\ \vdots \\ x[N-1] \end{bmatrix}}_{\mathbf{x}} = \underbrace{\begin{bmatrix} h[0] & 0 & \ldots & 0 \\ h[1] & h[0] & \ldots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ h[N-1] & h[N-2] & \ldots & h[N-n_s] \end{bmatrix}}_{\mathbf{H}} \underbrace{\begin{bmatrix} s[0] \\ s[1] \\ \vdots \\ s[n_s-1] \end{bmatrix}}_{\boldsymbol{\theta}} + \underbrace{\begin{bmatrix} w[0] \\ w[1] \\ \vdots \\ w[N-1] \end{bmatrix}}_{\mathbf{w}} \tag{11.38}$$

Note that in $\mathbf{H}$ the elements $h[n] = 0$ for $n > n_h - 1$. This is exactly in the form of the Bayesian linear model with $p = n_s$. Also, $\mathbf{H}$ is lower triangular due to the causal nature of $h(t)$ and thus $h[n]$. We need only specify the mean and covariance of $\boldsymbol{\theta} = \mathbf{s}$ to complete the Bayesian linear model description. If the signal is zero mean (as in speech, for example), we can assume the prior PDF

$$\mathbf{s} \sim \mathcal{N}(\mathbf{0}, \mathbf{C}_s).$$

An explicit form for the covariance matrix can be found by assuming the signal is WSS, at least over a short time interval (as in speech, for example). Then,

$$[\mathbf{C}_s]_{ij} = r_{ss}[i-j]$$

where $r_{ss}[k]$ is the ACF. If the PSD is known, then the ACF may be found and therefore $\mathbf{C}_s$ specified. As a result, the MMSE estimator of $\mathbf{s}$ is from (11.32)

$$\hat{\mathbf{s}} = \mathbf{C}_s\mathbf{H}^T\left(\mathbf{H}\mathbf{C}_s\mathbf{H}^T + \sigma^2\mathbf{I}\right)^{-1}\mathbf{x}. \tag{11.39}$$

An interesting special case occurs when $\mathbf{H} = \mathbf{I}$, where $\mathbf{I}$ is the $n_s \times n_s$ identity matrix. In this case the channel impulse response is $h[n] = \delta[n]$, so that the Bayesian linear model becomes

$$\mathbf{x} = \boldsymbol{\theta} + \mathbf{w}.$$

In effect the channel is transparent. Note that in the classical linear model we always assumed that $\mathbf{H}$ was $N \times p$ with $N > p$. This was important from a practical viewpoint since for $\mathbf{H} = \mathbf{I}$ the MVU estimator is

$$\begin{aligned} \hat{\boldsymbol{\theta}} &= (\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T\mathbf{x} \\ &= \mathbf{x}. \end{aligned}$$

In this case there is no averaging because we are estimating as many parameters as observations. In the Bayesian case we can let $N = p$ or even $N < p$ since we have prior knowledge about $\boldsymbol{\theta}$. If $N < p$, (11.39) still exists, while in classical estimation, the MVU estimator does not. We now assume that $\mathbf{H} = \mathbf{I}$. The MMSE estimator of $\mathbf{s}$ is

$$\hat{\mathbf{s}} = \mathbf{C}_s(\mathbf{C}_s + \sigma^2\mathbf{I})^{-1}\mathbf{x}. \tag{11.40}$$

Note that the estimator may be written as $\hat{\mathbf{s}} = \mathbf{A}\mathbf{x}$, where $\mathbf{A}$ is an $n_s \times n_s$ matrix. The $\mathbf{A}$ matrix is called the *Wiener filter*. We will have more to say about Wiener filters in Chapter 12. As an example, for the scalar case in which we estimate $s[0]$ based on $x[0]$ the Wiener filter becomes

$$\begin{aligned} \hat{s}[0] &= \frac{r_{ss}[0]}{r_{ss}[0] + \sigma^2}x[0] \\ &= \frac{\eta}{\eta + 1}x[0] \end{aligned}$$

where $\eta = r_{ss}[0]/\sigma^2$ is the SNR. For a high SNR we have $\hat{s}[0] \approx x[0]$, while for a low SNR $\hat{s}[0] \approx 0$. A more interesting example can be obtained by assuming that $s[n]$ is a realization of an AR(1) process or

$$s[n] = -a[1]s[n-1] + u[n]$$

where $u[n]$ is WGN with variance σ_u^2 and $a[1]$ is the filter coefficient. The ACF for this process can be shown to be (see Appendix 1)

$$r_{ss}[k] = \frac{\sigma_u^2}{1 - a^2[1]}\left(-a[1]\right)^{|k|}$$

and the PSD is

$$P_{ss}(f) = \frac{\sigma_u^2}{|1 + a[1]\exp(-j2\pi f)|^2}.$$

For $a[1] < 0$ the PSD is that of a low-pass process, an example of which is shown in Figure 11.9 for $a[1] = -0.95$ and $\sigma_u^2 = 1$. For the same AR parameters a realization of $s[n]$ is shown in Figure 11.10a as the dashed curve. When WGN is added to yield an

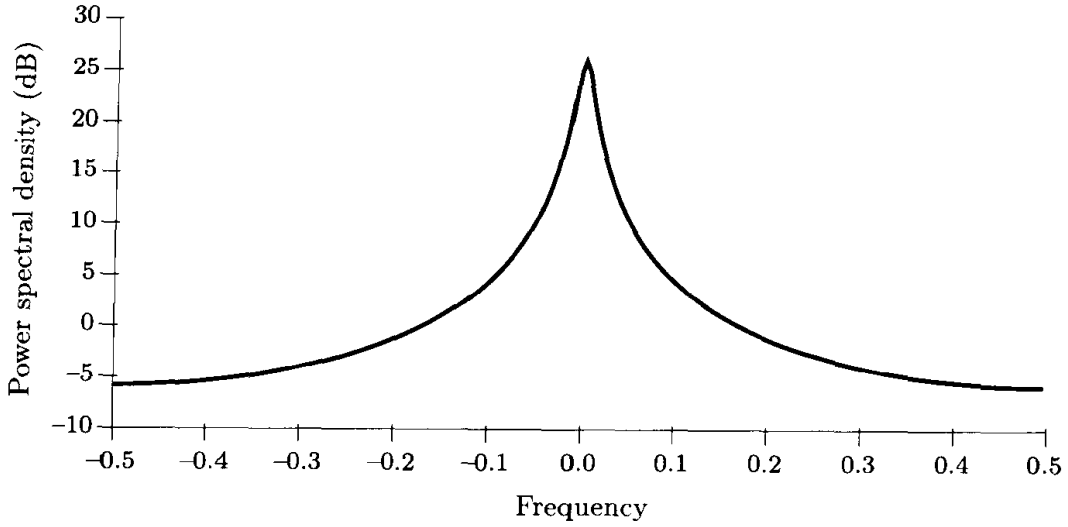


Figure 11.9 Power spectral density of AR(1) process

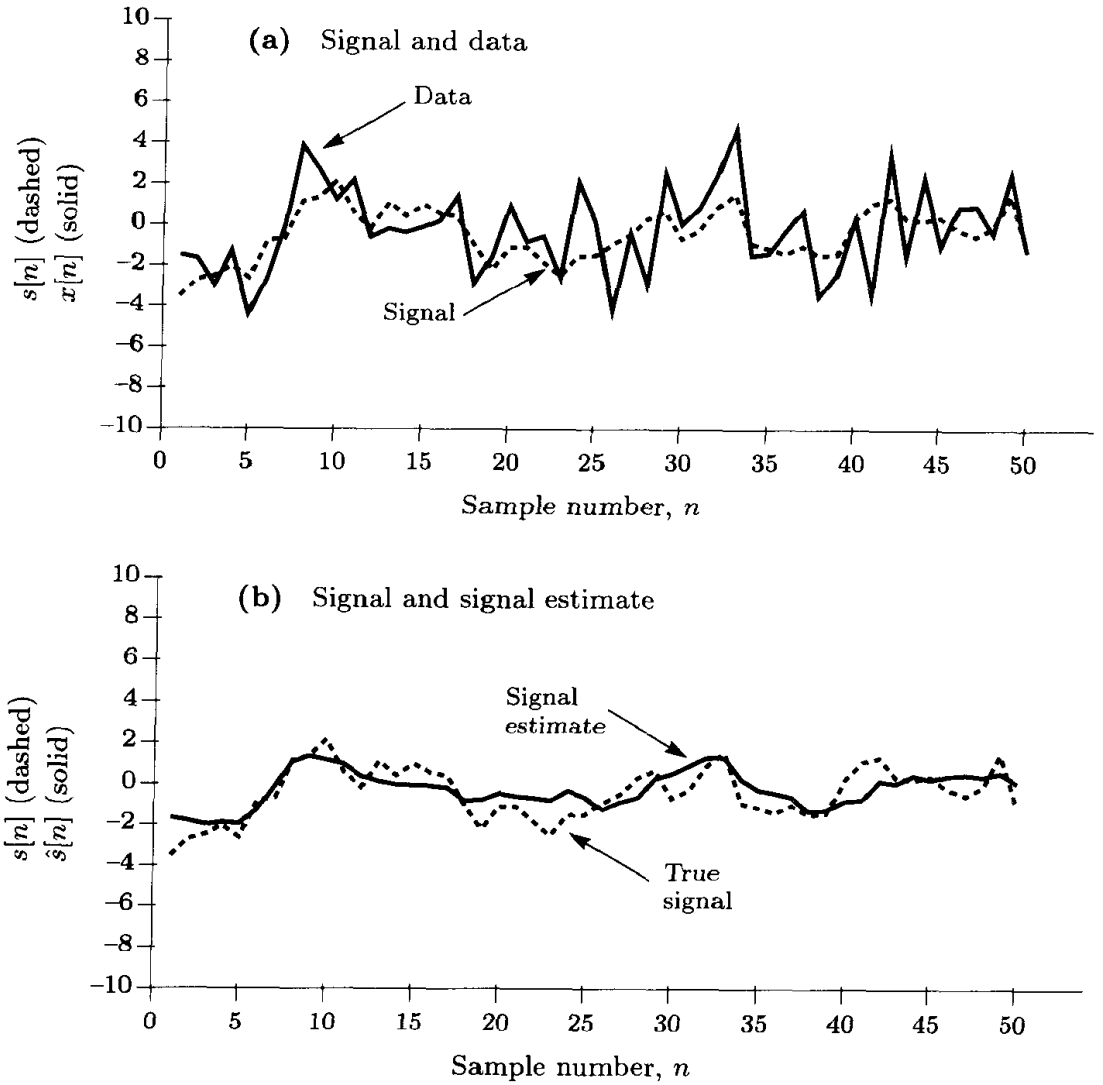


Figure 11.10 Wiener filtering example

SNR of 5 dB, the data $x[n]$ become the solid curve shown in Figure 11.10a. The Wiener filter given by (11.40) will smooth the noise fluctuations, as shown in Figure 11.10b by the solid curve. However, the price paid is that the signal will suffer some smoothing as well. This is a typical tradeoff. Finally, as might be expected, it can be shown that the Wiener filter acts as a low-pass filter (see Problem 11.18).

References

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Robinson, E.A, S. Treitel, *Geophysical Signal Analysis*, Prentice-Hall, Englewood Cliffs, N.J., 1980.
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Problems

11.1 The data $x[n]$, $n = 0, 1, \ldots, N-1$ are observed having the PDF

$$p(x[n]|\mu) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{1}{2\sigma^2}(x[n]-\mu)^2\right].$$

The $x[n]$'s are independent when conditioned on μ. The mean μ has the prior PDF

$$\mu \sim \mathcal{N}(\mu_0, \sigma_0^2).$$

Find the MMSE and MAP estimators of μ. What happens as $\sigma_0^2 \to 0$ and $\sigma_0^2 \to \infty$?

11.2 For the posterior PDF

$$p(\theta|x) = \frac{\epsilon}{\sqrt{2\pi}} \exp\left[-\frac{1}{2}(\theta-x)^2\right] + \frac{1-\epsilon}{\sqrt{2\pi}} \exp\left[-\frac{1}{2}(\theta+x)^2\right].$$

Plot the PDF for $\epsilon = 1/2$ and $\epsilon = 3/4$. Next, find the MMSE and MAP estimators for the same values of ϵ.

11.3 For the posterior PDF

$$p(\theta|x) = \begin{cases} \exp[-(\theta-x)] & \theta > x \\ 0 & \theta < x \end{cases}$$

find the MMSE and MAP estimators.

11.4 The data $x[n] = A + w[n]$ for $n = 0, 1, \ldots, N-1$ are observed. The unknown parameter A is assumed to have the prior PDF

$$p(A) = \begin{cases} \lambda \exp(-\lambda A) & A > 0 \\ 0 & A < 0 \end{cases}$$

where $\lambda > 0$, and $w[n]$ is WGN with variance σ^2 and is independent of A. Find the MAP estimator of A.

11.5 If $\boldsymbol{\theta}$ and $\mathbf{x}$ are jointly Gaussian so that

$$\begin{bmatrix} \boldsymbol{\theta} \\ \mathbf{x} \end{bmatrix} \sim \mathcal{N}(\boldsymbol{\mu}, \mathbf{C})$$

where

$$\begin{aligned} \boldsymbol{\mu} &= \begin{bmatrix} E(\boldsymbol{\theta}) \\ E(\mathbf{x}) \end{bmatrix} \\ \mathbf{C} &= \begin{bmatrix} \mathbf{C}_{\theta\theta} & \mathbf{C}_{\theta x} \\ \mathbf{C}_{x\theta} & \mathbf{C}_{xx} \end{bmatrix} \end{aligned}$$

find the MMSE estimator of $\boldsymbol{\theta}$ based on $\mathbf{x}$. Also, determine the minimum Bayesian MSE for each component of $\boldsymbol{\theta}$. Explain what happens if $\mathbf{C}_{\theta x} = \mathbf{0}$.

11.6 Consider the data model in Example 11.1 for a single sinusoid in WGN. Rewrite the model as

$$x[n] = A\cos(2\pi f_0 n + \phi) + w[n]$$

where

$$\begin{aligned} A &= \sqrt{a^2 + b^2} \\ \phi &= \arctan\left(\frac{-b}{a}\right). \end{aligned}$$

If $\boldsymbol{\theta} = [a\, b]^T \sim \mathcal{N}(\mathbf{0}, \sigma_\theta^2 \mathbf{I})$, show that the PDF of A is Rayleigh, the PDF of ϕ is $\mathcal{U}[0, 2\pi]$, and that A and ϕ are independent.

11.7 In the classical general linear model the MVU estimator is also efficient and therefore the maximum likelihood method produces it or the MVU estimator is found by maximizing

$$p(\mathbf{x}; \boldsymbol{\theta}) = \frac{1}{(2\pi)^{\frac{N}{2}} \det^{\frac{1}{2}}(\mathbf{C})} \exp\left[-\frac{1}{2}(\mathbf{x} - \mathbf{H}\boldsymbol{\theta})^T \mathbf{C}^{-1}(\mathbf{x} - \mathbf{H}\boldsymbol{\theta})\right].$$

For the Bayesian linear model the MMSE estimator is identical to the MAP estimator. Argue why in the absence of prior information the MMSE estimator for the Bayesian linear model is identical in form to the MVU estimator for the classical general linear model.

11.8 Consider a parameter $\boldsymbol{\theta}$ which changes with time according to the deterministic relation

$$\boldsymbol{\theta}[n] = \mathbf{A}\boldsymbol{\theta}[n-1] \qquad n \geq 1$$

where $\mathbf{A}$ is a known $p \times p$ invertible matrix and $\boldsymbol{\theta}[0]$ is an unknown parameter which is modeled as a random vector. Note that once $\boldsymbol{\theta}[0]$ is specified, so is $\boldsymbol{\theta}[n]$ for $n \geq 1$. Prove that the MMSE estimator of $\boldsymbol{\theta}[n]$ is

$$\hat{\boldsymbol{\theta}}[n] = \mathbf{A}^n \hat{\boldsymbol{\theta}}[0]$$

where $\hat{\boldsymbol{\theta}}[0]$ is the MMSE estimator of $\boldsymbol{\theta}[0]$ or, equivalently,

$$\hat{\boldsymbol{\theta}}[n] = \mathbf{A}\hat{\boldsymbol{\theta}}[n-1].$$

11.9 A vehicle starts at an unknown position $(x[0], y[0])$ and moves at a constant velocity according to

$$\begin{aligned} x[n] &= x[0] + v_x n \\ y[n] &= y[0] + v_y n \end{aligned}$$

where v_x, v_y are the velocity components in the x and y directions, respectively, and are both unknown. It is desired to estimate the position of the vehicle for each time n, as well as the velocity. Although the initial position $(x[0], y[0])$ and velocity $[v_x\, v_y]^T$ are unknown, they can be modeled by a random vector. Show that $\boldsymbol{\theta}[n] = [x[n]\, y[n]\, v_x\, v_y]^T$ can be estimated by the MMSE estimator

$$\hat{\boldsymbol{\theta}}[n] = \mathbf{A}^n\hat{\boldsymbol{\theta}}[0]$$

where $\hat{\boldsymbol{\theta}}[0]$ is the MMSE estimator of $[x[0]\, y[0]\, v_x\, v_y]^T$. Also, determine $\mathbf{A}$.

11.10 Show that the MAP estimator in Example 11.2 is consistent in the Bayesian sense or as $N \to \infty$, $\hat{\theta} \to \theta$ for any realization θ of the unknown parameter. Give an argument why this may be true in general. Hint: See Example 9.4.

11.11 Consider the vector MAP estimator or

$$\hat{\boldsymbol{\theta}} = \arg\max_{\boldsymbol{\theta}} p(\boldsymbol{\theta}|\mathbf{x}).$$

Show that this estimator minimizes the Bayes risk for the cost function

$$C(\boldsymbol{\epsilon}) = \begin{cases} 1 & ||\boldsymbol{\epsilon}|| > \delta \\ 0 & ||\boldsymbol{\epsilon}|| < \delta \end{cases}$$

where $\boldsymbol{\epsilon} = \boldsymbol{\theta} - \hat{\boldsymbol{\theta}}$, $||\boldsymbol{\epsilon}||^2 = \sum_{i=1}^{p} \epsilon_i^2$, and $\delta \to 0$.

11.12 Show that the MAP estimator of

$$\boldsymbol{\alpha} = \mathbf{A}\boldsymbol{\theta}$$

is

$$\hat{\boldsymbol{\alpha}} = \mathbf{A}\hat{\boldsymbol{\theta}}$$

where $\boldsymbol{\alpha}$ is a $p \times 1$ vector and $\mathbf{A}$ is an invertible $p \times p$ matrix. In other words, the MAP estimator commutes over invertible *linear* transformations.

11.13 If $\mathbf{x}$ is a 2×1 random vector with PDF

$$\mathbf{x} \sim \mathcal{N}(\mathbf{0}, \mathbf{C}),$$

prove that the PDF of $\mathbf{x}^T\mathbf{C}^{-1}\mathbf{x}$ is that of a χ_2^2 random variable. Hint: Note that $\mathbf{C}^{-1}$ may be factored as $\mathbf{D}^T\mathbf{D}$, where $\mathbf{D}$ is a whitening transformation as described in Section 4.5.

11.14 If $\boldsymbol{\epsilon}$ is a 2×1 random vector and $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{M}_{\hat{\theta}})$, plot the error ellipse for $P = 0.9$ for the following cases:

a. $\mathbf{M}_{\hat{\theta}} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

b. $\mathbf{M}_{\hat{\theta}} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$

c. $\mathbf{M}_{\hat{\theta}} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$.

11.15 As a special case of the Bayesian linear model assume that

$$\mathbf{x} = \boldsymbol{\theta} + \mathbf{w}$$

where $\boldsymbol{\theta}$ is a 2×1 random vector and

$$\begin{aligned} \mathbf{C}_w &= \sigma^2 \mathbf{I} \\ \mathbf{C}_\theta &= \begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{bmatrix}. \end{aligned}$$

Plot the error ellipse if $\sigma_1^2 = \sigma_2^2$ and $\sigma_2^2 > \sigma_1^2$. What happens if $\sigma_2^2 \gg \sigma_1^2$?

11.16 In fitting a line through experimental data we assume the model

$$x[n] = A + Bn + w[n] \qquad -M \leq n \leq M$$

where $w[n]$ is WGN with variance σ^2. If we have some prior knowledge of the slope B and intercept A such as

$$\begin{bmatrix} A \\ B \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} A_0 \\ B_0 \end{bmatrix}, \begin{bmatrix} \sigma_A^2 & 0 \\ 0 & \sigma_B^2 \end{bmatrix} \right)$$

find the MMSE estimator of A and B as well as the minimum Bayesian MSE. Assume that A, B are independent of $w[n]$. Which parameter will benefit most from the prior knowledge?

11.17 If $s[n] = A + Bn$ for $n = -M, -(M-1), \ldots, M$ and $\hat{A}, \hat{B}$ are the MMSE estimators for A and B, respectively, find the MMSE estimator $\hat{\mathbf{s}}$ of $\mathbf{s} = [s[-M]\, s[-(M-1)] \ldots s[M]]^T$. Next, find the PDF of the error $\boldsymbol{\epsilon} = \mathbf{s} - \hat{\mathbf{s}}$ if the data model is that given in Problem 11.16. How does the error vary with n and why?

11.18 In this problem we derive the noncausal Wiener filter for estimation of a signal in WGN. We do so by extending the results in Section 11.7. There the MMSE estimator of the signal was given by (11.40). First, show that

$$(\mathbf{C}_s + \sigma^2 \mathbf{I})\hat{\mathbf{s}} = \mathbf{C}_s \mathbf{x}.$$

Next, assume the signal is WSS, so that

$$\begin{aligned}[\mathbf{C}_s + \sigma^2\mathbf{I}]_{ij} &= r_{ss}[i-j] + \sigma^2\delta[i-j] = r_{xx}[i-j] \\ [\mathbf{C}_s]_{ij} &= r_{ss}[i-j].\end{aligned}$$

As a result, the equations to be solved for $\hat{\mathbf{s}}$ can be written as

$$\sum_{j=0}^{N-1} r_{xx}[i-j]\hat{s}[j] = \sum_{j=0}^{N-1} r_{ss}[i-j]x[j]$$

for $i = 0, 1, \ldots, N-1$. Next, extend this result to the case where $s[n]$ is to be estimated for $|n| \leq M$ based on $x[n]$ for $|n| \leq M$, and let $M \to \infty$ to obtain

$$\sum_{j=-\infty}^{\infty} r_{xx}[i-j]\hat{s}[j] = \sum_{j=-\infty}^{\infty} r_{ss}[i-j]x[j]$$

for $-\infty < i < \infty$. Using Fourier transforms, find the frequency response of the Wiener filter $H(f)$ or

$$H(f) = \frac{\hat{S}(f)}{X(f)}.$$

(You may assume that the Fourier transforms of $x[n]$ and $\hat{s}[n]$ exist.) Explain your results.

Appendix 11A

Conversion of Continuous-Time System to Discrete-Time System

A continuous-time signal $s(t)$ is the input to a linear time invariant system with impulse response $h(t)$. The output $s_o(t)$ is

$$s_o(t) = \int_{-\infty}^{\infty} h(t-\tau)s(\tau)\,d\tau.$$

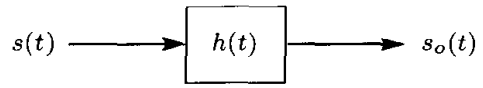
We assume that $s(t)$ is bandlimited to B Hz. The output signal must therefore also be bandlimited to B Hz and hence we can assume that the frequency response of the system or $\mathcal{F}\{h(t)\}$ is also bandlimited to B Hz. As a result, the original system shown in Figure 11A.1a may be replaced by the one shown in Figure 11A.1b, where the operator shown in the dashed box is a sampler and low-pass filter. The sampling function is

$$p(t) = \sum_{n=-\infty}^{\infty} \delta(t - n\Delta)$$

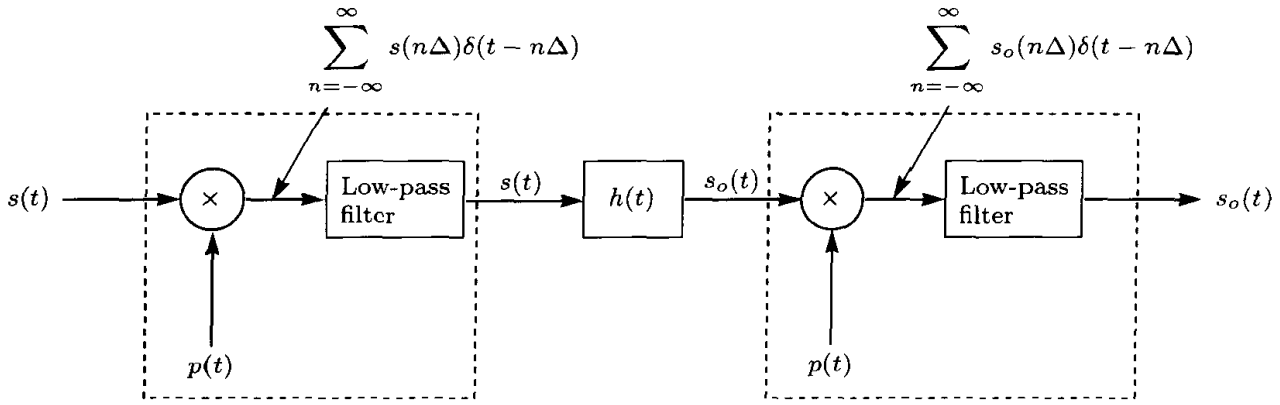
and the low-pass filter has the frequency response

$$H_{\text{lpf}}(F) = \begin{cases} \Delta & |F| < B \\ 0 & |F| > B. \end{cases}$$

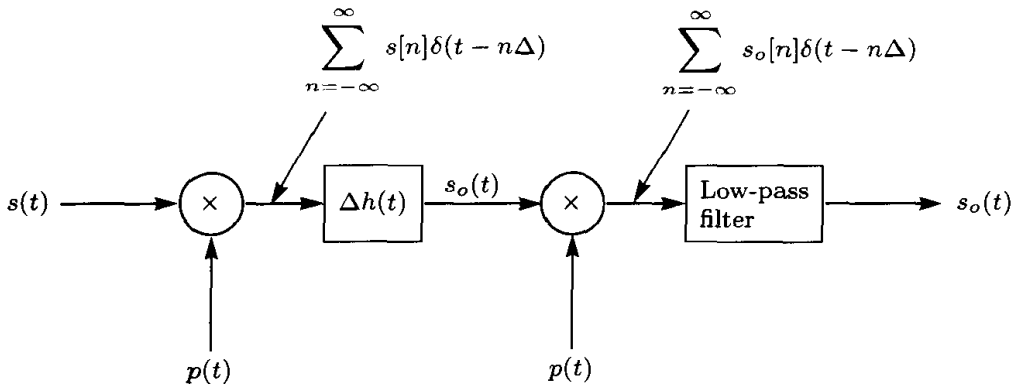
For bandlimited signals with bandwidth B the system within the dashed box does not change the signal. This is because the sampling operation creates replicas of the spectrum of $s(t)$, while the low-pass filter retains only the original spectrum scaled by Δ to cancel the $1/\Delta$ factor introduced by sampling. Next, note that the first low-pass filter can be combined with the system since $h(t)$ is also assumed to be bandlimited to B Hz. The result is shown in Figure 11A.1c. To reconstruct $s_o(t)$ we need only the



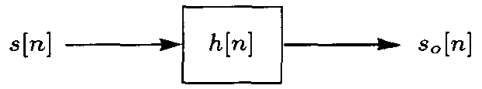
(a) Original continuous-time system



(b) Conversion to sampled data system



(c) Sampled data system



$$h[n] = \Delta h(n\Delta)$$

(d) Discrete-time system

Figure 11A.1 Conversion of continuous-time system to discrete-time system

samples $s_o(n\Delta)$. Referring to Figures 11.11b and 11.11c, we have

$$\begin{aligned} s_o(t) &= \int_{-\infty}^{\infty} \Delta h(t-\tau) \sum_{m=-\infty}^{\infty} s(m\Delta)\delta(\tau - m\Delta)\, d\tau \\ &= \Delta \sum_{m=-\infty}^{\infty} s(m\Delta) \int_{-\infty}^{\infty} h(t-\tau)\delta(\tau - m\Delta)\, d\tau \\ &= \Delta \sum_{m=-\infty}^{\infty} s(m\Delta) h(t - m\Delta) \end{aligned}$$

so that the output samples are

$$s_o(n\Delta) = \sum_{m=-\infty}^{\infty} \Delta h(n\Delta - m\Delta) s(m\Delta)$$

or letting $s_o[n] = s_o(n\Delta)$, $h[n] = \Delta h(n\Delta)$, and $s[n] = s(n\Delta)$, this becomes

$$s_o[n] = \sum_{m=-\infty}^{\infty} h[n-m]s[m].$$

The continuous-time output signal is obtained by passing the signal

$$\sum_{n=-\infty}^{\infty} s_o[n]\delta(t - n\Delta)$$

through the low-pass filter as shown in Figure 11A.1c. In any practical system we will sample the continuous-time signal $s_o(t)$ to produce our data set $s_o[n]$. Hence, we may replace the continuous-time data $s_o(t)$ by the discrete-time data $s_o[n]$, as shown in Figure 11A.1d, without a loss of information. Note finally that in reality $s(t)$ cannot be bandlimited since it is time-limited. We generally assume, and it is borne out in practice, that the signal is *approximately* bandlimited. The loss of information is usually negligible.